Keep, Measure, Erase

What a Quantum Loop Does with Its History

ANONYMOUS AUTHOR(S)

Quantum loops have a reputation for paradox. We trace the paradoxes to a single fact: the history of a quantum loop, the record of how many times its body ran, is itself quantum information. In a small quantum language in the Π family we can leave this history implicit, keep it, or measure it. The coherent loop is the direct-sum trace of its body: for every finite unitary body it is total and unitary and compiles exactly into loop-free code over a field with decidable equality. A lossless optical network realizes this trace in steady light of one frequency when each return path delays the wave by a whole number of cycles. Keeping and measuring use two universal constructions on quantum processes: $\mathscr{R}$ adds allocation for a finite counter, and $\mathscr{L}$ adds hiding so we can discard a fresh copy of the guard. Classical feedback then repeats the call on the returning outcome. Measuring the guard on every round gives the same data as keeping the exact number of returns and then forgetting it. Forgetting gives the coherent data exactly when the same count state factors out on every input. A finite test on the loop body decides when this holds, so a compiler can safely replace a coherent loop by a measured one in any channel context, even on entangled inputs. Averaging the coherent loop's output density matrix over a uniformly random return phase yields the measured data, and averaging the circulating weight gives the measured loop's mean number of returns. We implement the language with all three loops and run the paper's examples in it, using a floating-point interpreter and exact checks on real rational tables.

1 Introduction

Can a quantum computer run a loop? The question sounds settled, yet the answers on record seem to contradict one another. Let the loop's guard be in superposition, so that one branch exits after three rounds while another keeps going. If a clock records when each branch halts, branches that leave different records do not interfere [38, 42]. A second approach says the loop must measure its guard on every round, which leaves a classical loop running over quantum data [59]. The measurement supplies a Boolean condition, as in Q#'s `repeat ... until` or OpenQASM 3's `while` [18, 41, 44]. A third keeps the guard coherent but bounds the rounds and pads shorter branches until all reach the same control state at the same step [62, 63]. A fourth defines the loop as an infinite sum over the paths of its control flow [6]. Andrés-Martínez compares the clock's record with this path sum [3, Introduction]. In our approach, the differences arise from a single design choice, namely what to do with the iteration count:

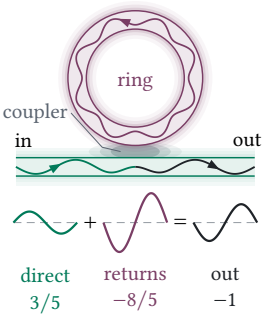
- **Keep it.** Return the count with the data, modulo K for a register of size K.
- **Measure it.** Measure the guard after each call to decide whether to continue.
- **Erase it.** Never write the count, so that the exit amplitudes add at the output.

For classical programs the three coincide on the returned data when the count is ignored, which is why classical languages never had to choose. For quantum programs they give three different loop constructs, and a language that provides only one has chosen for its programmers. We make the choice explicit in our language, so a programmer can write the loop they want.

Why should the count matter when nobody reads it? A quantum loop can exit on its third call and on its fifth, in superposition, and both exits can reach the same output. There the two contributions combine as complex amplitudes, which behave like signed weights: they can reinforce or cancel, so the final answer depends on both. A counter tags each contribution with its iteration number, and contributions with different tags stay apart whether or not anyone reads the tags. As in Bohr's two-slit example, leaving out the device that records the path can change the answer [10, Fig. 3]. For a reversible programmer the count is garbage that still carries information: if each round leaves a different tag, discarding the tags does not erase them: it gives the same data as measuring the guard on every round. Untagged, the contributions of all iteration counts add into a single map, the

infinite sum of the fourth answer. Category theory calls this map the direct-sum trace of the loop body [6, 35], and the geometry of interaction computes it with its execution formula [1].

Leaving the count implicit may look exotic, yet optical devices do it routinely. A ring resonator is a closed loop of optical waveguide, much like a ring of optical fiber, placed beside a straight waveguide (Inset 1). Where the two pass close together, light arriving on the straight guide either continues along it or crosses into the ring, and light already in the ring either goes around again or crosses back out. The waves overlap across the gap, so light can pass between the guides. This junction, called a coupler, plays the role of a loop body with an exit and a way back. Light that leaves after one trip, after two or after a hundred all reaches the same output. Under a steady beam of a single color, these contributions add as amplitudes [2, Eq. (6)] [3, § 5.3], just like the exits of a loop that leaves its count implicit. In our example the straight amplitude is $3/5$ and each crossing contributes $4i/5$. Crossing twice gives the returns their minus sign: together they contribute $-8/5$ to the direct wave's $3/5$, giving -1 (Section 5).



Inset 1. The ring coupler.

For branching in superposition, Bădescu and Panangaden suggest looking at optical devices such as the Mach–Zehnder interferometer [5, § 5], and we do: Figure 6 puts our loop on one of its paths.

2 Control and termination of quantum loops

Most quantum programs follow Selinger's slogan of quantum data with classical control [49, § 1]: a loop measures its guard and terminates as a probabilistic loop does [59, Def. 3.1]. Quantum control instead lets the choice of branch depend on data in superposition. Bădescu and Panangaden show that such a choice has no compositional meaning on channels, since branches that differ by a global phase are one channel yet give different controlled gates, and they find it seemingly incompatible with recursion [5, §§ 2, 3, 5].

For loops the proposals differ. Yuan et al. prove that the classical conditional jump cannot be lifted to superposition, with or without a jump history, and conclude that a machine with a program counter can leave the number of iterations in superposition only below a classical bound [62, §§ 1.2, 1.3]. In their bounded loop the count is an input, which tells each branch how often to repeat an operation. Their padding assumes a common upper bound: shorter branches execute no-ops until all reach the same control state at the same step [62, § 6.1, Fig. 7]. Reading the clock then reveals nothing about the count, which is still in the input. Register-machine compilation likewise computes waits within a time bound [63, § 5]. Ying, Yu and Feng instead record the history as the loop runs: every round uses a fresh coin [60, § 6], so that in Ying's loop k returns and an exit spell 1^k0 [58, Eq. (34)], and Assolini and Di Pierro likewise copy each test into a fresh qubit [4, § 4].

All of these loops treat termination classically: every branch ends at a definite round, which an outcome reveals, a bound fixes in advance, or fresh qubits record. The alternative is the infinite sum over the paths of a loop's control flow [6], in which no round is the last. Andrés-Martínez gives a new proof that this sum is well defined and a body on which it differs from the measured loop [3, Thm. 3.3.15, § 3.3.3]. Yet he judges this reading of a program not physically sound, because different paths would interact as if their outputs appeared at one instant. To avoid this he adds a delay, so that only exits at the same time step interfere [3, Introduction, § 5.1], and when every return takes one step the time of exit again marks the round. For plane waves in place of particles, he notes, feedback needs no delay, since such a wave is in a sense everywhere at every instant [3,

§ 5.3]. We work with such waves: in steady light of one frequency, all trips around a ring reach the output together.

To ask when a loop terminates is to ask for a count, so the question has an answer only for a loop that records one. We therefore sort loops by whether they keep, measure or erase the count.

We write the three loops in ISO, a small quantum language based on reversible pattern matching [14, 48]. We work with explicit tables, one column per basis input (2^q for q qubits), so the unit of compilation is a small body such as the ring's junction. To close it, `trace` feeds the state straight back, count_K also counts the returns modulo K, and `while` measures the guard after each call. The exits on the third and fifth calls have made two and four returns. If we start a bit at zero and flip it on each return, both exits leave it at zero and can still interfere. Measuring the guard on the third call tells them apart, since one exits and the other returns.

- **Coherent feedback compiles to a unitary gate.** We compile Bartha's finite unitary trace [6, Thm. 8] to a loop-free clause table by eliminating one return value at a time, following Andrés-Martínez [3, proof of Thm. 3.3.15] (Section 4).
- **Optics already builds the coherent loop.** For a unitary whose matrix is known, a lossless optical network in steady light realizes the same trace, whose physical interpretation Fairbanks and Selinger leave open [20, § 1, App. B] (Section 5).
- **Keeping and measuring come from two universal constructions.** James and Sabry recover irreversible classical programs from reversible ones by effects that create and erase information [33], and Heunen and Kaarsgaard recover measurement from unitaries in the same way, leaving iteration open [30, § 7]. Allocation supplies the counter and hiding lets the caller discard it [30, 31]. Classical feedback on the returning measurement outcome gives `while`, with the same data as discarding an exact count, and this loop terminates with probability one (Section 6).
- **Forgetting can be certified.** With m returning basis values and a nonempty input, we check whether the first $m+1$ exit maps are scalar multiples of one nonzero exit. The measured and coherent loops are then interchangeable in every channel context (Section 7).
- **The coherent loop can be probed without a count register.** Comparing two nearby return phases with one probe qubit estimates how much weight circulates in the loop, and averaging that weight over all phases gives the measured loop's mean number of returns (Section 8).

3 Finite programs in ISO

An iso $\omega : a \leftrightarrow b$ is a reversible program between finite types. In Π, 0 has no values, 1 has just $*$, $a + b$ holds a tagged value, and $a \otimes b$ holds a pair [30, § 3.1]. Each basis value $s \in \mathscr{B}(a)$ gives a vector $|s\rangle$. In $1 + 1$ only the tags distinguish the two copies of $*$.

3.1 Syntax

A clause $v \leftrightarrow e$ binds the variables of v in e. For instance, left x matches left s and binds x to s.

Each variable occurs only once in a pattern. On the right of a clause, brackets group a finite superposition. We build isos from clause tables and earlier definitions: $\oplus$ chooses a program by its input tag, $\otimes$ acts on both parts of a pair, and dagger reverses the iso. To run two isos in sequence on a closed term t, we write `f; g on t`, which passes the output of `f` to `g`.

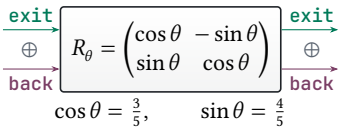
$$
\begin{aligned}
v &::= * \mid x \mid \mathsf{left}\, v \mid \mathsf{right}\, v \mid \langle v_1, v_2 \rangle, \\
p, q &::= * \mid x \mid \langle p, q \rangle, \\
e &::= v \mid \textstyle\sum_i \alpha_i e_i \mid \mathsf{let}\ p = \omega\, q\ \mathsf{in}\ e, \\
\mathscr{T} &::= \{v_i \leftrightarrow e_i\}_{i \in I}, \qquad t ::= e \mid \omega\, t, \\
\omega &::= f \mid \mathscr{T} \mid \omega; \omega \mid \omega \oplus \omega \mid \omega \otimes \omega \mid \omega^\dagger.
\end{aligned}
$$

A clause pattern v can test a tag. In a local call q assembles the argument and p binds the result for e. Since one continuation accepts every basis output, its pattern uses only unit, pairs and variables.

An ordinary definition pairs a signature `f :: a <-> b` with a body `f = omega`. The declaration `round` $f : a + c = \mathcal{T}$ gives f type $(a + c) \leftrightarrow (a + c)$, with the same external type a on both sides. The outer tag is the guard that selects the port: `exit` (left) carries an external value of a, and `back` (right) a return of c. We write `Unit` for 1 and `unit` for $*$. At `Qubit`, or $1 + 1$, `0` and `1` stand for left $*$ and right $*$. Thus `exit 0` expands to left (left $*$), with one tag for the port and one for its payload.

For our first round both ports carry `Unit`. One call rotates the guard through $\theta = \arccos(3/5)$ in the ordered basis `exit unit`, `back unit`.

```
round guardRound : Unit + Unit = {
  exit unit ↔ [ 3/5 . exit unit + 4/5 . back unit];
  back unit ↔ [-4/5 . exit unit + 3/5 . back unit]
}
```



On `exit unit` the dots scale the two outputs, giving $(3/5)|\text{exit } *\rangle + (4/5)|\text{back } *\rangle$. This is the first column of R_θ, and `back unit` supplies the second. The loop will connect the return ports.

Unfolding the names in an iso and replacing each checked table by its matrix atom gives a *matrix combinator*. For `guardRound` the atom is $[R_\theta]$.

$$\kappa ::= [M] \mid \kappa;\kappa \mid \kappa \oplus \kappa \mid \kappa \otimes \kappa \mid \kappa^\dagger.$$

3.2 Statics

To check a table $\omega : a \leftrightarrow b$, we first check that each basis input of a matches exactly one clause and that each output expression has type b under the variables its pattern binds. On basis input s, the clause gives column s of the table's matrix M. Its row t is the amplitude for output t. The marked entry takes `back` to `exit`.

	exit	back
exit	3/5	−4/5
back	4/5	3/5

For `guardRound` both squared column norms are one. In the inner product the exit row contributes $-12/25$ and the back row $12/25$ (Appendix Figure 13, in the supplementary material). In general we check that M is square and $M^\dagger M = I$, where dagger takes the conjugate transpose. This tests every squared norm on the diagonal and every pair of different columns off it.

The judgment $\Sigma; \Delta \vdash e : b$ gives e output type b in the earlier iso definitions Σ and the context of value variables Δ. In a pair pattern $\langle v_1, v_2 \rangle$, linear use splits Δ between the components. A superposition offers alternatives, so each alternative uses the same variables once. A local call splits variables between argument and continuation, where its result pattern binds fresh variables.

The judgment $\Sigma \vdash \omega : a \leftrightarrow b$ checks tables and matrix atoms by the same column test, which is decidable over fields such as $\mathbb{Q}(i, \sqrt{2})$. For a composite we check that the interfaces match. Checking a table evaluates its clauses, but a named call uses a shorter declaration prefix and an inline call a smaller subprogram. Checking and evaluation therefore terminate by induction in lexicographic order (Appendix A) [48, §§ 2, 5] [37, Ch. 3].

3.3 Dynamics

To compute a column of M, we match a basis input against its clause. With the resulting bindings γ, a value v gives $|v[\gamma]\rangle$ and a sum gives the weighted sum of its vectors. A local `let` uses each basis output of the called iso to bind the continuation's variables, then adds the resulting vectors with their amplitudes. We write $\Sigma; \gamma \vdash e \Downarrow \psi$ for its output vector under the declarations Σ.

For `guardRound`, elaboration gives the atom $[R_\theta]$ and evaluation returns its matrix R_θ. In general, if elaboration gives $\mathcal{E}_\Sigma(\omega) = \kappa$ and $\kappa \Downarrow M$, we write $\Sigma \vdash \omega \Downarrow M$. Applying the program to a closed input $\Sigma \vdash t \Downarrow \psi$ then gives $\Sigma \vdash \omega\, t \Downarrow M\psi$.

On `exit unit` the first call to `guardRound` gives both tags. A second call weights its two columns by the first call's amplitudes.

$$\frac{3}{5}\begin{pmatrix}3/5\\4/5\end{pmatrix}+\frac{4}{5}\begin{pmatrix}-4/5\\3/5\end{pmatrix}=\begin{pmatrix}-7/25\\24/25\end{pmatrix}.$$

The second call receives the whole output of the first, so $\kappa \Downarrow M$ and $\lambda \Downarrow N$ give $\kappa;\lambda \Downarrow NM$. Sum, tensor and dagger take the direct sum, tensor product and conjugate transpose of these matrices.

3.4 Denotation

A type a denotes $[\![a]\!] = \mathbb{C}^{\mathscr{B}(a)}$. Thus 0 denotes the zero space and 1 denotes $\mathbb{C}$, with direct sum and tensor for sums and products. The clauses give the columns of $[\![\omega]\!] : [\![a]\!] \to [\![b]\!]$. For `guardRound` linearity therefore gives the rotation shown here.

$$R_\theta\begin{pmatrix}u\\v\end{pmatrix}=u\begin{pmatrix}3/5\\4/5\end{pmatrix}+v\begin{pmatrix}-4/5\\3/5\end{pmatrix}=\begin{pmatrix}(3u-4v)/5\\(4u+3v)/5\end{pmatrix}.$$

The column check makes this map unitary (Appendix A).

4 The coherent loop as a sum of exits

In sequence, both outputs of `guardRound` enter the next call. When we close the return wire, only the lower one goes around. The upper output sends $3/5$ to the caller, where later exits add to it.

4.1 From a round to its exits

Follow the $4/5$ on the lower wire in Figure 1. The second call sends $(4/5)(-4/5) = -16/25$ to the upper port and $(4/5)(3/5) = 12/25$ back for a third call. A classical reversible body follows one path on each basis input. Here several exits reach the same port. We number the trips to work out what each contributes, but the program puts no trip number on the output. With nothing there to distinguish the exits, only their sum reaches the caller.

Syntax and statics. Elaboration turns `trace guardRound` into $\mathrm{trace}_1[R_\theta]$. Both rules close the same wire, one on source clauses and one on their matrix:

$$\underbrace{\frac{\Sigma \vdash \omega : a+c \leftrightarrow b+c}{\Sigma \vdash \mathrm{trace}_c\,\omega : a \leftrightarrow b}}_{\text{source iso}} \qquad \underbrace{\frac{\kappa : a+c \leftrightarrow b+c}{\mathrm{trace}_c\,\kappa : a \leftrightarrow b}}_{\text{matrix combinator}}.$$

Elaboration preserves this connection: $\mathscr{C}_\Sigma(\mathrm{trace}_c\,\omega) = \mathrm{trace}_c\,\mathscr{C}_\Sigma(\omega)$. In `trace f; g` we close `f` first. The body is unitary and has the same return type c on both sides, so its external dimensions agree. A `round` also uses the same external type. This is the do-while interface of qWhile [3, § 5.1], using Bartha's trace on finite types as proposed by Lemonnier [37, § 5.2.3].

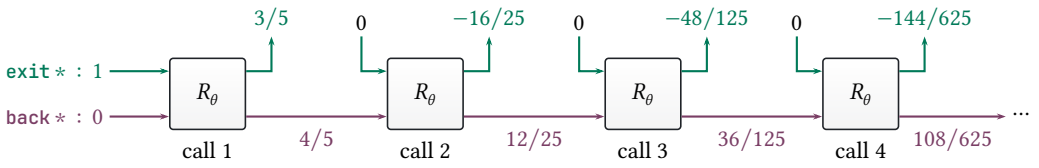


Fig. 1. Four calls to `guardRound`. The upper outputs go to the caller: the first two add to $-1/25$ and all of them to $3/5 - 16/25 - 48/125 - 144/625 - \cdots = -1$. The lower wire returns amplitude to the next call, whose external input is zero.

In Figure 2 the gates are maps in Unitary, the category of finite-dimensional Hilbert spaces and unitary maps. The guard factorization $F_g : P \oplus P \to \mathbb{C}^2 \otimes P$ puts both alternatives on one sheet.

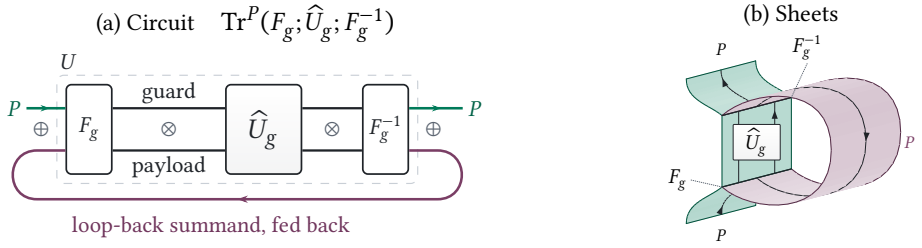


Fig. 2. For a common payload P, read the circuit rightward and the sheets upward. Separate sheets carry alternatives ($\oplus$), and wires on one sheet carry pairs ($\otimes$) [17, Defs. 16, 18–20, 22]. The middle box acts by $\widehat{U}_g = F_g U F_g^{-1}$. After F_g^{-1} separates the ports, closing the return gives the direct-sum trace $P \to P$ [6, 35].

The exit maps. For a body $\beta : a + c \leftrightarrow b + c$, write $X = [\![a]\!]$, $X' = [\![b]\!]$ and $Y = [\![c]\!]$. Group the columns of its matrix U by input tag and the rows by output tag. On external input u and return y,

$$U\begin{pmatrix} u \\ y \end{pmatrix} = \begin{pmatrix} A & B \\ C & D \end{pmatrix}\begin{pmatrix} u \\ y \end{pmatrix} = \begin{pmatrix} Au + By \\ Cu + Dy \end{pmatrix}. \tag{4.1}$$

For `guardRound` these columns are $(3/5, 4/5)$ and $(-4/5, 3/5)$. In **FdHilb**, which has the same spaces and all linear maps, the four blocks have types $A : X \to X'$, $B : Y \to X'$, $C : X \to Y$ and $D : Y \to Y$. Both sides have $m = \dim Y$ return values, leaving $n = \dim X = \dim X'$ external ones.

The first call sends Au outside and Cu back. The next takes $(0, Cu)$ to (BCu, DCu). After $k + 1$ returns, amplitude has entered through C, passed through D another k times and left through B. Writing $x_k : X \to X'$ for the exit after k returns gives

$$x_0 = A, \qquad x_{k+1} = BD^k C. \tag{4.2}$$

After $k + 1$ calls, $x_0 u, \ldots, x_k u$ have reached the external output and $D^k C u$ remains on the wire. Adding all the exits gives the execution formula $[\![\mathrm{trace}_c\, \beta]\!] = L = \sum_{k \geq 0} x_k$ [1, 22]. In the ring, x_0 gives the direct path and x_k the light that made k trips.

The only returns that can circulate forever without emitting are in $Y_0 = \bigcap_{j \geq 0} \ker BD^j$. A unitary body carries $0 \oplus Y_0$ onto itself, since every state there returns with its full norm. An external input $(u, 0)$ lies in the orthogonal complement, so its returns stay in $Y_1 = Y_0^\perp$, where the powers of D decay geometrically (Appendix B). This is why all the incoming probability eventually exits.

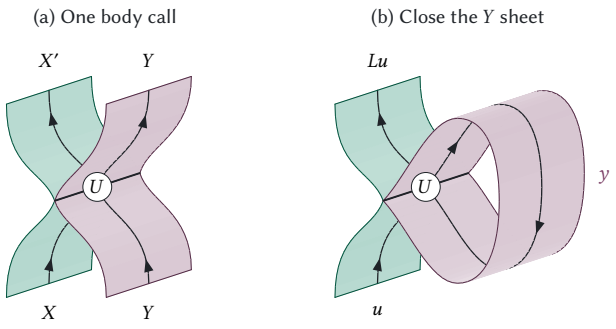
4.2 Computing the sum without a last round

On input `unit`, `guardRound` sends $4/5$ back on its first call and $3/5$ of what it receives on each later call (Figure 1). If y is the sum of the returns, the terms after the first sum to $(3/5)y$.

$$y = \frac{4}{5} + \underbrace{\frac{12}{25} + \frac{36}{125} + \frac{108}{625} + \frac{324}{3125} + \cdots}_{(3/5)y} = \frac{4}{5} + \frac{3}{5}y.$$

This is the *wire equation* for `guardRound`. In general, Cu first enters the return wire and each later call applies D, so the sum satisfies $y = Cu + DCu + D^2Cu + \cdots = Cu + Dy$. The caller receives the immediate exit Au together with By, as Figure 3 illustrates.

Theorem 4.1 (Finite feedback). *For a finite unitary body, Y_0 and $Y_1 = Y_0^\perp$ reduce D, C maps into Y_1, and the powers of $D|_{Y_1}$ decay geometrically. The exits sum absolutely to a unitary L and satisfy $\sum_k x_k^\dagger x_k = I_X$. Every input u admits a solution of $y = Cu + Dy$, and each solution gives $Lu = Au + By$.*



Both ends of the return sheet carry the same amplitude y. For $u = 1$ the wire equation gives

$$\tfrac{2}{5}y = \tfrac{4}{5}, \qquad y = 2.$$

The external output receives $3/5$ directly and $-4/5$ of each return, giving $3/5 - (4/5)y = -1$. Thus

$$R_\theta \underbrace{\begin{pmatrix}1\\2\end{pmatrix}}_{(u,y)} = \underbrace{\begin{pmatrix}-1\\2\end{pmatrix}}_{(Lu,y)}.$$

Fig. 3. In Unitary, the open body at left is $U : X \oplus Y \to X' \oplus Y$. Joining the two ends of the purple Y sheet at right gives its direct-sum trace $L = \mathrm{Tr}^Y(U) : X \to X'$.

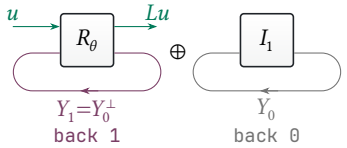
The first call splits $\|u\|^2$ into $\|Au\|^2 + \|Cu\|^2$. The next splits $\|Cu\|^2$ into $\|BCu\|^2 + \|DCu\|^2$. Since the remaining weight tends to zero, the squared exit norms sum to $\|u\|^2$. Closing the wire gives $U(u, y) = (Lu, y)$, so canceling $\|y\|^2$ shows $\|Lu\| = \|u\|$. Equal external dimensions make L unitary, as in Bartha's finite trace theorem [6, Thms. 4, 7–8]. Appendix B proves the decay bound.

Let column s of F sum the returns of input $|s\rangle$. The wire equations read $(I_Y - D)F = C$, and column s of $A + BF$ is the output.

$$\frac{\kappa \Downarrow U \quad U = \left(\begin{smallmatrix} A & B \\ C & D \end{smallmatrix}\right) \quad F : X \to Y \quad (I_Y - D)F = C}{\mathrm{trace}_c\, \kappa \Downarrow A + BF}.$$

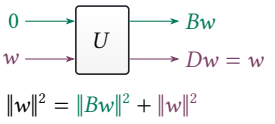
An undetermined return. For `guardRound` we divided by $1 - 3/5$. To see why $I_Y - D$ can have no inverse, keep the original return on `back 1` and let a second value, `back 0`, return to itself.

```
round extraPort : Unit + Qubit = {
  exit unit ↔ [ 3/5 . exit unit + 4/5 . back 1];
  back 1    ↔ [-4/5 . exit unit + 3/5 . back 1];
  back 0    ↔ back 0
}
```



Here `back 0` spans the trapped space Y_0 of Theorem 4.1, and `back 1` spans its orthogonal complement $Y_1 = Y_0^\perp$. The rotation gives $y_1 = \frac{4}{5}u + \frac{3}{5}y_1$, so $y_1 = 2u$. The identity gives $y_0 = y_0$ for every y_0, so $I_Y - D = \mathrm{diag}(2/5, 0)$ in the order (`back 1`, `back 0`) has no inverse. Starting from u gives $y_0 = 0$ because no call reaches `back 0`. Only `back 1` feeds the exit, so every choice of y_0 gives $(3/5)u - (4/5)y_1 = -u$.

For input u, the sum of the returns gives one solution y. If $y + w$ is another, the outputs differ by Bw. Subtracting the wire equations cancels the common first return Cu, leaving $Dw = w$. As the lower inset shows, one call returns the whole difference unchanged. Since U preserves norm, nothing remains at the exit ($Bw = 0$), so both solutions give the caller $Au + By$.



So far our wire has fed amplitude from one call to the next. Czachor uses the same connection to model a wormhole, feeding one port's later output back to its earlier input [19, Eqs. (6)–(7)]. The grandfather paradox imagines a traveler who goes back and prevents their own departure. Czachor models that encounter at the entrance, where the incoming wave meets the wave that has gone around the loop. For a lossless loop with $I_Y - D$ invertible, adding the amplitudes of all trips gives a unitary map on the outside port. To a traveler who tries to enter, it is as if the universe had placed

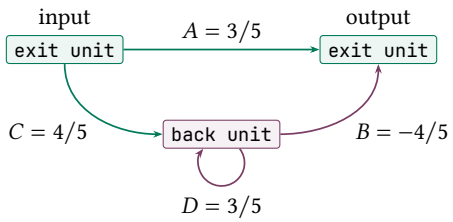
a mirror across the doorway: all the probability eventually comes back to the outside. A perfect absorber inside the wormhole cuts off the wave that made the entrance act like a mirror. If the entrance sends some of the incoming wave into the loop, the traveler now has a chance to step through. If they do, it is a one-way ticket: the absorber stops them before they can complete the trip [19, Thm. 1 and § IV].

Czachor calls this reflection *chronology protection* for a wormhole that already exists [19, § I]. Hawking and Visser ask the earlier question of what prevents one from forming [28, 53].

When a single complex amplitude returns unchanged, Czachor observes that its port is isolated, just like `back 0`. He asks what happens with more return ports [19, arXiv v8, after Eq. (45)]. Bartha's work on quantum Turing automata gives the finite answer: he proved that the paths to the exit sum to an isometry for any finite isometric body, even if $I_Y - D$ is singular [6, Eq. (1), Thm. 8]. Here $Bw = 0$ shows why the caller gets the same output from any solution.

4.3 Compiling away the loop

Building on Bartha's proof, Andrés-Martínez eliminates one return value at a time for finite contractions [3, Thm. 3.3.15, Cor. 3.3.16, comparison on pp. 165–166]. We use that construction to write the clauses of $L = A + BF$. In `guardRound` the first clause sends $3/5$ to `exit unit` and $4/5$ to `back unit`. To replace `back unit` in that clause, we sum the exits from that port.



On input `back unit`, each further return contributes a factor D. The exits sum to

$$B + BD + BD^2 + \cdots = \frac{B}{1 - D} = -2.$$

The first clause sends $C = 4/5$ there, giving $(4/5)(-2)$. The wire equation groups the same factors as $BF = (-4/5) \cdot 2$, with $F = C/(1 - D)$.

In the first clause we replace `back unit` by $-2\,(\texttt{exit unit})$. Both terms now leave by the same port, so we can add their coefficients and drop the external tag.

$$\texttt{exit unit} \leftrightarrow \tfrac{3}{5}\,(\texttt{exit unit}) + \tfrac{4}{5}\big(-2\,(\texttt{exit unit})\big) = -\texttt{exit unit} \qquad \rightsquigarrow \qquad \{* \leftrightarrow -1 \cdot *\}.$$

In `flipRound`, each call rotates the guard and flips a data bit. Both clauses call `X`, which swaps `0` and `1`, before forming their outputs (Appendix F.1).

```
round flipRound : Qubit + Qubit = {
  exit x ↔ let y = X x in [ 3/5 . exit y + 4/5 . back y];
  back x ↔ let y = X x in [-4/5 . exit y + 3/5 . back y]
}
```

$$U_{\text{flip}} = R_\theta \otimes \mathsf{X}$$

Neither `back 0` nor `back 1` therefore returns to itself in one call. Follow `back 1` in Figure 4. It can return to `back 0` and then to `back 1`, with amplitude $3/5$ each way. Removing `back 0` turns those two steps into a single return with amplitude $9/25$. We therefore divide the remaining exits of `back 1` by $1 - 9/25$ before substituting them in the external clauses.

For a general body, we evaluate each basis input as in Section 3 and choose a return **back** q. Write the evaluated clause as **back** $q \mapsto d\,(\textbf{back}\,q) + e$, where d is the self-entry of q in the current table and e contains the other outputs. If $d \neq 1$, we replace each occurrence of $\alpha\,(\textbf{back}\,q)$ in the other clauses by $\alpha e/(1 - d)$, collect equal patterns and delete the clause for q. If $|d| = 1$, its row and column already have full norm: $e = 0$ and no other clause feeds q. In particular when $d = 1$ we can delete it without dividing, as in `extraPort`.

One call

$$\texttt{exit}\,0 \mapsto \tfrac{3}{5}\texttt{exit}\,1 + \tfrac{4}{5}\texttt{back}\,1 \qquad \texttt{exit}\,1 \mapsto \tfrac{3}{5}\texttt{exit}\,0 + \tfrac{4}{5}\underline{\texttt{back}\,0}$$

$$\texttt{back}\,0 \mapsto -\tfrac{4}{5}\texttt{exit}\,1 + \tfrac{3}{5}\texttt{back}\,1 \qquad \texttt{back}\,1 \mapsto -\tfrac{4}{5}\texttt{exit}\,0 + \tfrac{3}{5}\underline{\texttt{back}\,0}$$

Remove `back 0`

$$\texttt{exit}\,1 \mapsto \tfrac{3}{5}\texttt{exit}\,0 + \tfrac{4}{5}\underline{\left(-\tfrac{4}{5}\texttt{exit}\,1 + \tfrac{3}{5}\texttt{back}\,1\right)} = \tfrac{3}{5}\texttt{exit}\,0 - \tfrac{16}{25}\texttt{exit}\,1 + \tfrac{12}{25}\texttt{back}\,1,$$

$$\texttt{back}\,1 \mapsto -\tfrac{4}{5}\texttt{exit}\,0 + \tfrac{3}{5}\underline{\left(-\tfrac{4}{5}\texttt{exit}\,1 + \tfrac{3}{5}\texttt{back}\,1\right)} = -\tfrac{4}{5}\texttt{exit}\,0 - \tfrac{12}{25}\texttt{exit}\,1 + \underline{\tfrac{9}{25}\texttt{back}\,1}.$$

Close `back 1`

$$\texttt{back}\,1 \mapsto \tfrac{25}{16}\left(-\tfrac{4}{5}\texttt{exit}\,0 - \tfrac{12}{25}\texttt{exit}\,1\right) = \underline{-\tfrac{5}{4}\texttt{exit}\,0 - \tfrac{3}{4}\texttt{exit}\,1}.$$

Both exits

$$\texttt{exit}\,0 \mapsto \tfrac{3}{5}\texttt{exit}\,1 + \tfrac{4}{5}\underline{\left(-\tfrac{5}{4}\texttt{exit}\,0 - \tfrac{3}{4}\texttt{exit}\,1\right)} = -\texttt{exit}\,0,$$

$$\texttt{exit}\,1 \mapsto \tfrac{3}{5}\texttt{exit}\,0 - \tfrac{16}{25}\texttt{exit}\,1 + \tfrac{12}{25}\underline{\left(-\tfrac{5}{4}\texttt{exit}\,0 - \tfrac{3}{4}\texttt{exit}\,1\right)} = -\texttt{exit}\,1.$$

Fig. 4. Compiling `trace flipRound`. The top row gives the four clauses of $R_\theta \otimes \mathsf{X}$, read from input to output. Underlining marks each substitution. The self-entry of `back 0` is zero. Closing `back 1` multiplies its other outputs by $(1 - 9/25)^{-1} = 25/16$. After these substitutions, both inputs leave unchanged with amplitude -1.

Proposition 4.2 (Compiling away the loop). *Let the evaluated clause table of a finite unitary body have coefficients in a subfield $\mathbb{F} \subseteq \mathbb{C}$ with decidable equality. Eliminating the $|\mathscr{B}(c)|$ return values in any order, one per step, gives a loop-free table over $\mathbb{F}$ denoting L.*

To recover the original wire equations we put back the value we substituted, choosing any value when $d = 1$. Each step traces out just $\mathbb{C}q$ with the other ports still open, so Theorem 4.1 gives another unitary table (Appendix B.1). Dropping the outer `exit` tags gives a loop-free table. In `kickbackExact`, for example, eliminating the returns gives $I \otimes \mathsf{X}$: the data bit flips even on inputs entangled with the ancilla (Appendix F.1).

Solving the scalar wire equation when $d \neq 1$ gives Mason's signal-flow reduction [40, §§ 3.1–3.2], also used for zero-delay feedback when $I - D$ is invertible [25, Thm. 12, Eq. (12)] and for optical networks [52, App. A, Eq. (A1)]. Each of the m steps updates at most $(n+m)^2$ entries, for $O(m(n+m)^2)$ field operations. The test $d = 1$ uses exact equality. A hardware gate set may still need an approximation because dividing by $1 - d$ can leave its coefficient ring.

Bends and crossings. The direct sum makes Unitary a traced monoidal category [3, Cor. 3.3.16]: its trace $L = \mathrm{Tr}^Y(U)$ closes the return wire in one step. To draw that same connection as two bends, we need a type for the wire that runs back, which Int(Unitary) supplies [35, § 4]. Following the bent wire gives $y = Cu + Dy$, and reading the boxes of Appendix Figure 23(b) from left to right gives the same gate we just compiled (Appendix G).

4.4 Making the sign observable

Both the identity and `trace guardRound` return `unit` with certainty. To see the sign we need a second amplitude for comparison. In Figure 5, `bypassRound` sends $|0\rangle$ straight through and $|1\rangle$ through the rotation, whose return still carries `Unit`.

```
round bypassRound : Qubit + Unit = {
  exit 0 ↔ exit 0;
  exit 1 ↔ [ 3/5 . exit 1 + 4/5 . back unit];
  back unit ↔ [-4/5 . exit 1 + 3/5 . back unit]
}
```

$$U_{\text{bypass}} = \left(\begin{array}{cc|c} 1 & 0 & 0 \\ 0 & t & -s \\ \hline 0 & s & t \end{array}\right)$$

With $t = 3/5$ and $s = 4/5$, both paths contribute to the immediate exit $x_0 = \mathrm{diag}(1, t)$. Only the rotation returns, so $x_{k+1} = \mathrm{diag}(0, -s^2 t^k)$ for $k \geq 0$. Closing the wire fixes $|0\rangle$ and sends $|1\rangle$ to $-|1\rangle$.

The Hadamard test [15, p. 1, Eqs. (1)–(3)] compares the two paths. In `H; trace bypassRound; H on 0`, the first H prepares $(|0\rangle + |1\rangle)/\sqrt{2}$ and the loop changes the plus to a minus. Both states give equally likely bits until the second H recombines them:

$$H\frac{|0\rangle - |1\rangle}{\sqrt{2}} = \tfrac{1}{2}(|0\rangle + |1\rangle) - \tfrac{1}{2}(|0\rangle - |1\rangle) = (\tfrac{1}{2} - \tfrac{1}{2})|0\rangle + (\tfrac{1}{2} + \tfrac{1}{2})|1\rangle = |1\rangle.$$

A measurement now reads 1 with certainty. With no sign change the Hadamards would give 0.

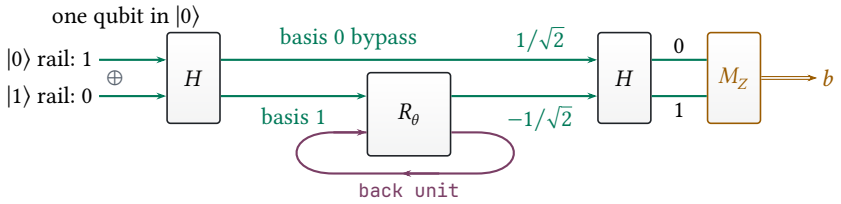


Fig. 5. The rails carry the basis alternatives of one qubit $G = \mathbb{C}^2$. The bypass beside the closed loop gives $\mathrm{diag}(1, -1) : G \to G$. Each Hadamard acts on both rails. Measurement gives one bit, here 1 with certainty.

5 The coherent loop in optics

Fairbanks and Selinger ask whether a device can take the trace of a unitary supplied as a black box [20, App. B]. Knowing the matrix, we can build the network and choose which ports to connect.

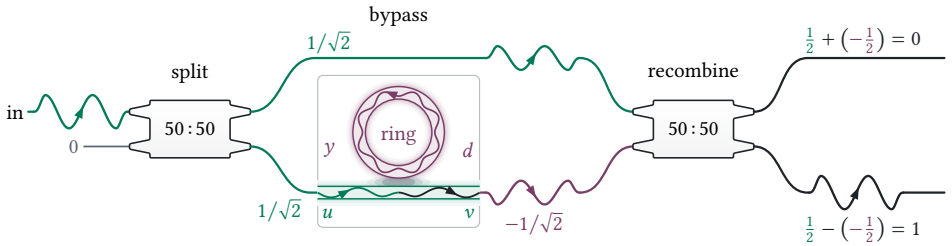
Proposition 5.1 (Optical feedback). *Realize a finite unitary body $U : X \oplus Y \to X' \oplus Y$ at the working frequency as a lossless optical network, one mode per basis value [47]. Connect each returning output to its matching input by a lossless delay of a whole number of wave periods. On steady input u at that frequency the output is Lu.*

Proof. At the working frequency each delay line returns its field unchanged. The steady amplitudes y on the returning inputs and v on the external outputs therefore satisfy $y = Cu + Dy$ and $v = Au + By$, the wire equations of Section 4. By Theorem 4.1, every solution gives $v = Lu$. □

For q qubits this construction uses 2^q optical modes, one path for each basis state of a single excitation. If the return ports are already accessible, we need only add the delays.

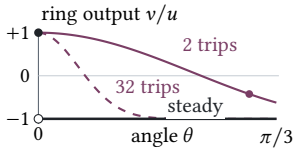
The splitters in Figure 6 form a Mach–Zehnder interferometer, and with their path phases they act like the Hadamards in Figure 5. When the ring reverses the sign, the two paths cancel at the sum port and all the light leaves by the difference port. At a frequency where a trip takes a whole number of cycles plus half a cycle (round-trip phase $z = -1$), the ring preserves the sign and the light leaves by the sum port. Wang et al. used a ring on the shorter path to send alternate frequencies in a regularly spaced sequence mostly to opposite outputs [54, Figs. 1 and 6].

Fairbanks and Selinger point out a difficulty for a device that takes the trace of a unitary. The rotation varies continuously with its angle, yet its trace is $+1$ at $\theta = 0$ and -1 for $0 < \theta < 2\pi$ [20, App. B, Remark B.1]. A fixed number of black-box calls gives a continuous response and cannot reproduce this jump (Appendix B). In the ring, the trace gives the amplitude we approach as light from more and more trips reaches the output. To follow that buildup, suppose we send a steady unit input into an empty ring. We choose the frequency so that one trip takes a whole number of wave cycles. At the coupler, a copy delayed by two trips then lines up crest for crest with one delayed by a single trip. This is resonance. The direct field $t = \cos\theta$ arrives first. After $n \geq 1$ trips the returning light brings the output to $t - (1 - t^2)(1 + t + \cdots + t^{n-1}) = -1 + (1 + t)t^n$.



$\binom{v}{d} = \begin{pmatrix} t & is \\ is & t \end{pmatrix}\binom{u}{y}, \quad y = zd$ `exit` $* \mapsto t$ `exit` $* + is$ `back` $*$ `back` $* \mapsto is$ `exit` $* + t$ `back` $*$

Fig. 6. The second splitter takes the normalized sum of the two paths at the upper port and their normalized difference at the lower. At the gap, light stays in its guide with amplitude $t = 3/5$ or crosses with $is = 4i/5$ [2, 46] (Appendix H). The guide carries u to v and the ring feeds d back as $y = zd$. At $z = 1$ the two crossings give $(is)^2 = -s^2$ and each further trip contributes t. The delayed exits of `guardRound` therefore sum to $-s^2/(1-t) = -8/5$. The plot varies $t = \cos\theta$. The steady output jumps at zero. The dot on the two-trip curve marks our $t = 3/5$.



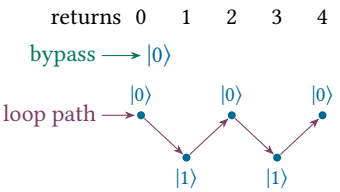
At $\theta = 0$ no light crosses the gap. For any fixed number of trips, the output approaches $+1$ as the angle approaches zero (Figure 6). If we hold a small positive angle fixed and wait, t^n decays to zero and the output approaches -1.

At resonance the returning amplitude has magnitude $|s|/(1-t)$ for $s \neq 0$, so the circulating weight $\cot^2(\theta/2)$ grows without bound near $\theta = 0$. With a frequency-independent coupler and a fixed round-trip time, the output phase changes with angular frequency at a rate equal to this weight times one round-trip time (Section 8). This is the Wigner–Smith delay [29, 50, 56].

A flash shorter than one trip behaves differently: the arrival time of each pulse records the count [46, Eqs. (30)–(33)]. In the program we can choose how much of that record to keep.

6 Counting and measured loops from allocation and hiding

In `bypassRound`, start a bit at 0 and flip it on each return. Exits on calls three and five have made two and four returns, so both carry the same 0 as the bypass. They can still interfere with it. An exit after an odd number of returns carries 1 and is distinguishable from all three.



The caller did not supply a count register. We need the ability to allocate that register.

Adding allocation to ISO. Our caller supplies a qubit with two basis values, but returning it alongside a counter bit gives four output pairs. Tagging the inputs gives a square table in ISO.

```
tagBit :: (Qubit + Qubit) ↔ (Qubit x Qubit)
tagBit = {left x  ↔ ⟨0, x⟩;
          right x ↔ ⟨1, x⟩}
lift tagBit on 1
```



The bit records which input tag to restore on reversal. With `lift`, u enters on the left, so the output is $|0\rangle \otimes u$, whether $u = |1\rangle$ as in the listing or $u = |+\rangle$ as in the caller.

Heunen and Kaarsgaard call their allocation language $\mathrm{U}\Pi_a$ ("yuppie-a") [30, § 3.3]. Our executable language $\textsc{iso}_a$ uses the same subscript. We write arr to embed an iso and lift to allocate.

$$\frac{\omega : a \leftrightarrow b}{\mathsf{arr}\,\omega : a \rightarrowtail b} \qquad \frac{\omega : a + d \leftrightarrow b}{\mathsf{lift}\,\omega : a \rightarrowtail b}.$$

For a body with matrix U, arr keeps the whole matrix and lift acts by $u \mapsto U(u, 0)$. The selected columns remain orthonormal, so this is an *isometry* ($\rightarrowtail$). Identifying extensions that agree on those columns gives $\mathscr{R}[\mathsf{Unitary}] \simeq \mathsf{Isometry}$ under sum and tensor [32, Prop. III.2, Thm. III.3] [30, Thm. 22]. In `count K f on t`, `f` is the round, `t` its input and $[K]$ the $K \geq 1$ counter values. We use alloc_0 to prepare zero and advance on every return. For $\beta : a + c \leftrightarrow b + c$ this gives $\mathsf{count}_K\,\beta : a \rightarrowtail [K] \otimes b$, with the counter first in formulas and the data first in code.

Adding hiding to $\textsc{iso}_a$. For the second Hadamard the caller must be able to hide the counter. In `bypassRound`, let u_0 sum the even exits, with the bypass, and u_1 the odd ones. The joint output $w = |0\rangle \otimes u_0 + |1\rangle \otimes u_1$ has density matrix $ww^\dagger$. Its partial trace groups rows and columns by the bit:

$$\begin{pmatrix} u_0u_0^\dagger & u_0u_1^\dagger \\ u_1u_0^\dagger & u_1u_1^\dagger \end{pmatrix} \xrightarrow{\text{hide the bit}} u_0u_0^\dagger + u_1u_1^\dagger.$$

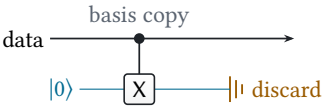
Only the diagonal blocks survive because $\langle 0 \mid 1\rangle = 0$. Within each block the exits carrying that bit have already added their amplitudes.

Heunen and Kaarsgaard add hiding in $\mathrm{U}\Pi_a^\chi$ ("yuppie-chi-a") [30, § 3.4]. Our $\textsc{iso}_a^\chi$ embeds each isometry with arr. We write hide for their second lift to name the output we discard.

$$\frac{v : a \rightarrowtail b}{\mathsf{arr}\,v : a \rightsquigarrow b} \qquad \frac{v : a \rightarrowtail b \otimes g \qquad \mathscr{B}(g) \neq \emptyset}{\mathsf{hide}\,v : a \rightsquigarrow b}.$$

The rule hides the right factor, so we swap first in $\mathsf{hide}(\mathsf{count}_K\,\beta; \mathsf{arr}\,\sigma_{[K],b})$. In code we can apply `discard` directly. On a density matrix ρ (a positive operator of trace one), arr v gives $V\rho V^\dagger$ and hide v gives $\mathrm{tr}_g(V\rho V^\dagger)$. Isometries on the hidden factor generate the equivalence in $\mathscr{L}[\mathscr{R}[\mathsf{Unitary}]]$ [30, Eq. (3), Thm. 22]. On isometries it agrees with Huot and Staton's affine reflection, giving CPTP, the completely positive trace-preserving maps, under composition and tensor [31, Cor. 7]. Hiding b in $b \leftrightarrow 1 \otimes b$ discards an inhabited b and gives $\mathrm{tr}\,\rho$ (Appendix Figure 14).

From a fresh bit to a measurement. Start a fresh bit at 0 and flip it when the data bit is 1. On basis input s this writes a copy of s. On a superposition $\alpha|0\rangle + \beta|1\rangle$, the two wires carry $\alpha|00\rangle + \beta|11\rangle$. Discarding the copy destroys the coherence between 0 and 1. Reading the data now gives 0 with probability $|\alpha|^2$ and 1 with probability $|\beta|^2$.



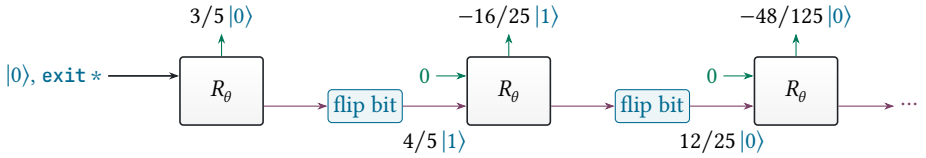
For inhabited b, lifting a permutation gives the basis copier $\mathsf{clone}_b : s \mapsto \langle s, s\rangle$. Hiding the copy measures b in that basis [30, § 3.4.1].

$$\mathsf{measure}_b = \mathsf{hide}\,\mathsf{clone}_b, \qquad [\![\mathsf{measure}_b]\!](\rho) = \sum_{s \in \mathscr{B}(b)} P_s \rho P_s, \qquad P_s = |s\rangle\langle s|.$$

In $\mathsf{while}_c\,\beta : a \rightsquigarrow b$ we copy just the tag, so hiding the copy preserves coherence within each payload (empty types are treated in Appendix A.5).

6.1 Keep a count

In `count 2 guardRound on unit`, the 2 tells us to keep the count modulo two. The first call sends 3/5 to the exit with the bit still at 0. The other 4/5 goes around with the bit flipped to 1. After two returns the bit is 0 again, so the first and third exits carry the same bit and payload.

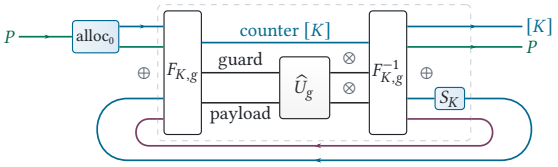


Every two returns multiply a delayed exit by $(3/5)^2$. The even tail sums to $-3/5$, canceling the immediate $3/5$. Only the odd exits survive.

$$\underbrace{\frac{3}{5} - \frac{48/125}{1-(3/5)^2}}_{\text{even returns}} = 0, \qquad \underbrace{\frac{-16/25}{1-(3/5)^2}}_{\text{odd returns}} = -1.$$

In `bypassRound` the loop path therefore leaves with $-|1\rangle$ in the counter and the bypass with $|0\rangle$. On input $|+\rangle$ the joint output is $(|0\rangle_c \otimes |0\rangle - |1\rangle_c \otimes |1\rangle)/\sqrt{2}$, with the counter first. Hiding it leaves the data in $I/2$, which the second Hadamard preserves, so the caller reads 1 half the time.

With K values, the bit flip becomes $S_K : j \mapsto j+1 \bmod K$. For a common payload P, $F_{K,g}$ distributes the counter over both ports and factors out the guard (Appendix C). The counter passes beside the body. After $F_{K,g}^{-1}$ restores the ports, the return carries both counter and payload. Only the counter passes through S_K.



Allocating zero and closing the enlarged body β_K gives exit $x_k u$ the label $k \bmod K$ (Figure 7):

$$\mathsf{count}_K\, \beta = \mathsf{alloc}_0\,;\, \mathsf{arr}(\mathsf{trace}_{[K]\otimes c}\, \beta_K), \qquad R_K u = \sum_{k\geq 0} |k \bmod K\rangle \otimes x_k u. \tag{6.1}$$

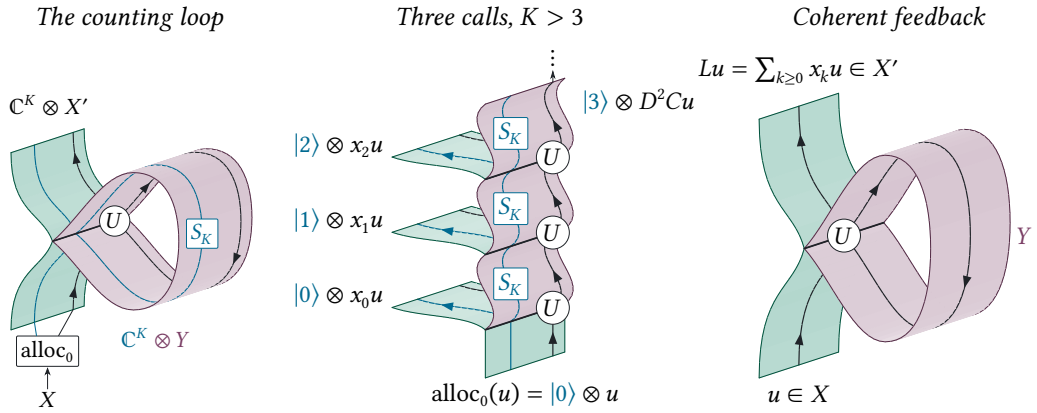


Fig. 7. Read upwards. Each trip around the folded sheet passes through S_K, advancing the counter. After three calls, $|3\rangle \otimes D^2Cu$ continues around the loop. Without the counter, all exit amplitudes add at one output.

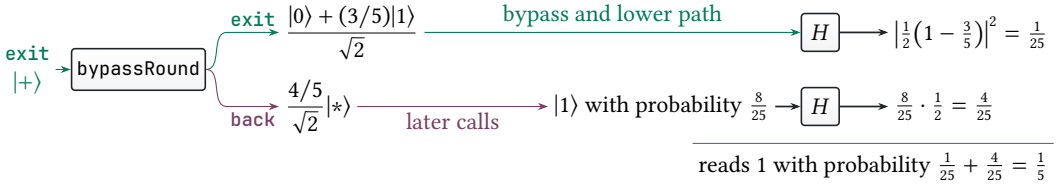
Discarding a written count does not erase it. The adjoint of the enlarged unitary in (6.1) restores $|0\rangle \otimes u$, after which a caller who knows the body can apply L. A Fourier measurement uses only the counter: the uniform outcome overlaps every label by $1/\sqrt{K}$, leaving $Lu/\sqrt{K}$ with probability $1/K$. Hiding the counter instead preserves interference within each residue. For `bypassRound` its probability tends to $1/5$ but differs from `while` at every finite K (Appendix C).

Allocating zero followed by a unitary makes R_K an isometry. The exact record $Ru = \sum_k |k\rangle \otimes x_k u$ in $\ell^2(\mathbb{N}) \otimes X'$ also preserves the norm, since $\sum_k \|x_k u\|^2 = \|u\|^2$. It lies outside the finite language, but the labels agree until the counter wraps. The remaining exits decay geometrically, so embedding

the first K labels gives operator-norm convergence $R_K \to R$ (Appendix C). Unlike a count supplied as an input (Section 2), our counter records when amplitude leaves, even when a single input gives exits at infinitely many counts.

6.2 Measure the returning tag

The caller now runs `H; while bypassRound; H; measure on 0`. The parity bit gave the first and third exits the same label. Measuring the tag on every call distinguishes them because one has already left when the other returns. Each path starts with amplitude $1/\sqrt{2}$, and we follow each tag of the first call to the caller's bit.



On `exit`, all of the bypass meets $3/5$ of the lower path. The second Hadamard subtracts the two amplitudes and divides by $\sqrt{2}$, giving amplitude $(1 - 3/5)/2 = 1/5$ at output 1. The bypass never returns, so `back` tells us that the qubit took the lower path. It returns again with probability $9/25$ at each later call, so it eventually exits as $|1\rangle$, and the second Hadamard sends half of its $8/25$ to output 1. In all, the caller reads 1 with probability $1/25 + 4/25 = 1/5$.

In the ring, a pulse shorter than one trip leaves as a train of pulses, one for each exit [46, Eqs. (30)–(33)]. We time the bypass to meet the direct pulse (Figure 8). Each later pulse reaches the splitter alone and splits equally, giving $1/5$ overall. This reminds us of Bergson's first book, where he observes that we can hear a bell's rhythm, yet to count its strokes we must keep them distinct [8, pp. 86–87]. The separate pulses do precisely this: their time bins are orthogonal, so a detector that ignores arrival time adds their probabilities.

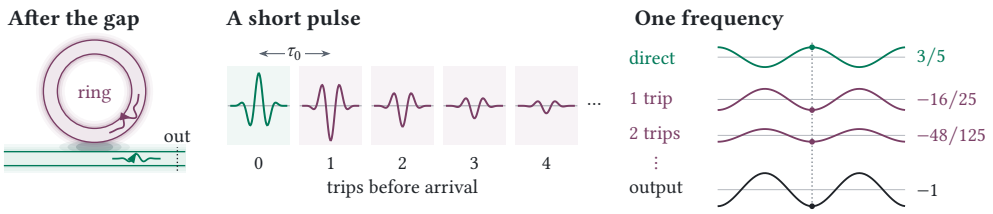


Fig. 8. Successive exits of a short pulse arrive one trip apart. For steady light at $z = 1$, the dots mark the amplitude of each wave at the same instant. They add to give the sign change in Figure 6.

What one call measures. After each call we copy the output tag of the iso and hide the copy (Figure 9). On the first exit both paths of `bypassRound` carry `exit`, so the copy leaves them free to interfere. A compiled `trace` can serve as this iso because it is unitary. A `while` returns a channel, so it cannot supply the body.

For input ρ and return σ, we keep the exit and return blocks.

$$\Phi_U(\rho,\sigma) = \big(\underbrace{A\rho A^\dagger + B\sigma B^\dagger}_{\text{exit}},\ \underbrace{C\rho C^\dagger + D\sigma D^\dagger}_{\text{return}}\big).$$

For a normalized input the block traces give the two probabilities. On $(\rho, 0)$ the first exit is $A\rho A^\dagger$ and the next is $(BC)\rho(BC)^\dagger$. Ying and Feng test the guard first [59], so when the external types agree we run U once before their loop. Each unfolding then adds one more of our exits, so Ying's least fixed point [57] gives their sum, also the trace over a classical sum [49, § 6.4] (Appendix C).

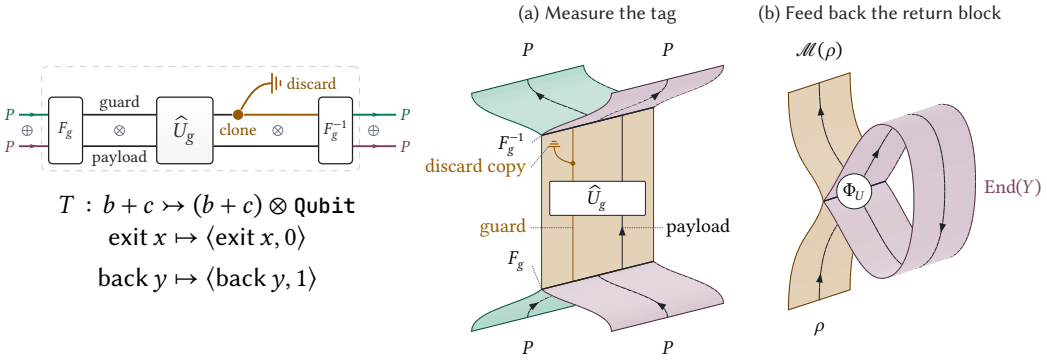


Fig. 9. One call is hide(arr β; T). With a common payload, F_g exposes the guard. Copying that qubit implements T in these coordinates, and discarding the copy dephases the two tags before F_g^{-1} restores the ports. Circuit and sheet (a) show one call with both ports open. In (b) the return block in End(Y) goes around again.

Adding the measured exits. On call $k+1$ an exit with k returns gives `exit` and every later exit gives `back`. For `guardRound`, the first two exits therefore contribute probability $(3/5)^2 + (-16/25)^2$, whereas coherent feedback adds their amplitudes to $-1/25$ (Appendix Table 2). Keeping all the tag outcomes and hiding them at the end gives the same sum of probabilities:

$$\mathscr{M}(\rho) = \sum_{k\geq 0} x_k \rho x_k^\dagger = \mathrm{tr}_{\ell^2(\mathbb{N})}(R\rho R^\dagger), \qquad \mathscr{M} = [\![\text{while}_c\ \beta]\!]. \tag{6.2}$$

Since $\sum_k x_k^\dagger x_k = I$ (Theorem 4.1), the probabilities sum to one even without a last call. A smaller register can realize the same channel without storing the count, as Appendix C shows for `bypassRound`.

Table 1. Erase, keep and measure, in row order, for `bypassRound` on $|+\rangle$ ($t = 3/5$). The first two rows give state vectors and the last a density matrix. Only the data reach the final H.

Loop	*General output*	*Bypass output*	Pr(1) after H
$\text{trace}_c\ \beta : a \leftrightarrow b$	$Lu = \sum_{k\geq 0} x_k u$	$\lvert - \rangle$	1
$\text{count}_K\ \beta : a \rightarrowtail [K] \otimes b$	$R_K u = \sum_{k\geq 0} \lvert k \bmod K \rangle \otimes x_k u$	$\frac{\lvert 0\rangle_c \otimes \lvert 0\rangle - \lvert 1\rangle_c \otimes \lvert 1\rangle}{\sqrt{2}}$	$1/2$ $(K = 2)$
$\text{while}_c\ \beta : a \rightsquigarrow b$	$\mathscr{M}(\rho) = \sum_{k\geq 0} x_k \rho x_k^\dagger$	$\frac{1}{2}\begin{pmatrix} 1 & t \\ t & 1 \end{pmatrix}$	$1/5$

In Table 1 the same caller reads 1 with certainty for coherent feedback, with probability $1/2$ when we keep parity, and with probability $1/5$ when we measure on every round.

7 Agreement of the measured and coherent loops

With the bypass, learning that the qubit returned even once tells us which path it took. In `guardRound` every exit carries `unit`, whatever the count. A compiler can choose either loop if both give the caller the same data on every input. To see when this works, keep the exact count for a moment.

Suppose each exit x_k applies the same gate L up to a scalar b_k. Keeping the count gives the joint output

$$Ru = \sum_k |k\rangle \otimes x_k u = \Big(\sum_k b_k |k\rangle\Big) \otimes Lu.$$

For `guardRound`, $L = -1$, so factoring out Lu leaves $-\frac{3}{5}|0\rangle + \frac{16}{25}|1\rangle + \frac{48}{125}|2\rangle + \cdots$ in the counter. Knowing how many returns occurred need reveal nothing about the data. Here the whole count

state, including its relative phases, is the same for every input, so the caller receives Lu even after discarding the count.

Theorem 7.1 (Agreement on the returned data). *Let $\beta : a + c \leftrightarrow b + c$ be a finite iso with nonempty external input type a. Let $m = |\mathscr{B}(c)|$ be the number of basis values on the return wire. The following are equivalent.*
(1) The loops agree on the returned data, $\mathscr{M}(\rho) = L\rho L^\dagger$ for every input state ρ.
(2) One unit count vector $\xi \in \ell^2(\mathbb{N})$ satisfies $Ru = \xi \otimes Lu$ for every input u.
(3) The first $m + 1$ exits span a line. One is nonzero and all the others in the list are its scalar multiples. In this case $x_k = b_k L$ for every k, where $\xi = \sum_k b_k |k\rangle$, $\sum_k b_k = 1$ and $\sum_k |b_k|^2 = 1$.

If the loops agree, any direction orthogonal to Lu has probability zero on unit input u. Each exit contributes a nonnegative probability there, so $x_k u$ must lie along Lu. For two distinct basis inputs e_i, e_j, applying this fact to $e_i + e_j$ and comparing the orthogonal coefficients along Le_i and Le_j forces their multipliers to agree. The same b_k therefore works for every basis input and, by linearity, $x_k = b_k L$. For finitely many exits this also follows from the freedom in a Kraus representation [55, Cor. 2.23]. Conversely, tracing out a fixed unit count vector leaves $L\rho L^\dagger$.

For $m > 0$, Cayley–Hamilton writes every D^j with $j \geq m$ in terms of $I, D, \ldots, D^{m-1}$. Multiplying by B and C puts every delayed exit in the span of $x_1, \ldots, x_m$. Including the immediate exit A gives a closed span containing the sum $L \neq 0$, even without a last call. When $m = 0$, only A remains. Since C lands in Y_1, we could use $d = \dim Y_1$ in place of m throughout. Using m spares us computing Y_1.

For the delayed exits $x_{k+1} = BD^kC$ we keep $W = D^kC$, starting at $W = C$: BW is the next exit and DW the next W. On dense tables a step costs $O(mn^2 + m^2n)$ field operations, so m steps cost $O(m^2n^2 + m^3n)$. We use a nonzero entry of the first nonzero exit P to test all the scalar ratios in $O((m + 1)n^2)$ comparisons. On success, any column of P has squared norm $q > 0$ and the channel is $\rho \mapsto P\rho P^\dagger/q$. Elimination recovers the gate with its phase. The ratios must be exact: near $t = 1$, rounding a small bypass exit to zero can accept a replacement that changes the caller's probability from $(1 - t)/2$ to one. A direct channel test instead solves for up to m^2 return entries (Appendix D).

The bound is sharp: the first m exits are not enough. Suppose $|0\rangle$ exits at once and $|1\rangle$ after $m > 0$ returns, both unchanged. Every exit between $x_0 = |0\rangle\langle 0|$ and $x_m = |1\rangle\langle 1|$ is zero, so only x_m reveals the second direction. A counter modulo m labels both exits zero and preserves $|+\rangle$, while measurement leaves $I/2$. This also attains the bound in Corollary 7.2 (Appendix D).

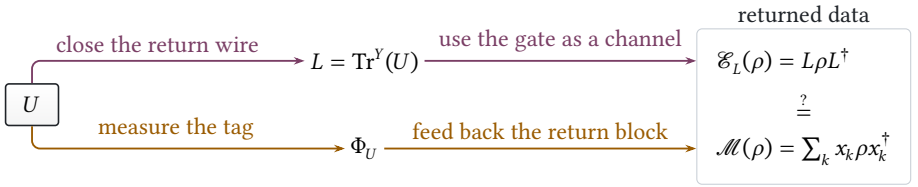


Fig. 10. Closing $U : X \oplus Y \to X' \oplus Y$ in Unitary gives $L : X \to X'$. Measuring the tag gives $\Phi_U : \mathrm{End}(X) \oplus \mathrm{End}(Y) \to \mathrm{End}(X') \oplus \mathrm{End}(Y)$, whose return we close by the classical-sum trace on completely positive trace-nonincreasing maps (Appendix C). Theorem 7.1 decides when the two channels agree.

Replacing the loop in a program. For `flipRound`, the exits $(3/5)\mathsf{X}$ and $-(16/25)I$ already distinguish the channels of Figure 10. In `flipOnExit` we flip the data only as it leaves. Its blocks are $(A, B, C, D) = (t\mathsf{X}, -s\mathsf{X}, sI, tI)$, so the immediate exit is $t\mathsf{X}$ and the exit after $k + 1$ returns is $-s^2t^k\mathsf{X}$. One X can therefore replace the measurements. In the other direction, repeating an available body may cost less than synthesizing the gate. Paetznick and Svore's retry resets the failed guard to zero and reuses the external input port. Feeding it into the return port instead changes the next exit

(Appendix D) [45, Fig. 9]. Our exact checker accepts six of the nine rational bodies in Appendix Table 3.

Leaving a hole in the caller where it uses the loop's data channel gives a *channel context*. The hole can follow, precede or run beside another channel, or occupy one branch of a finite measurement. Equality survives each operation, including a retained reference. For finitely many uses we replace one at a time, holding the others fixed (Appendix D). Unequal channels can be distinguished by an input and a measurement.

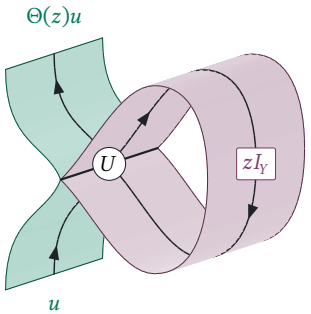
Corollary 7.2 (Replacement in a caller). *Hiding the exact count gives the measured loop. For a finite iso with nonempty external input, the measured and coherent loops agree in every channel context exactly when $x_0, \dots, x_m$ span a line. Then hiding the count modulo any $K \geq 1$ gives this channel too. Conversely, agreement of the hidden and coherent channels for one $K > \dim Y_1$ suffices. On empty input all these channels agree.*

Tensoring the closed channel with a reference is allowed in a context. In `bypassRound` we instead add an identity summand *before* closing the body. Only the immediate exit acts on this summand. Any delayed exit proportional to it must therefore vanish, so replacement with a bypass is safe exactly when every input exits at once.

8 Applications of the coherent loop known from physics

Physics has long used the loop that writes no count, in rings, cavities and scattering regions, and learns what circulates from its response to a phase on the return path. We do the same for a clause table. The identities are known, and we add a one-qubit program that reads the circulating weight.

A phase on the return wire. Once `guardRound` is closed, the caller can see the sign change by using a bypass. What the sign alone cannot tell us is how it arose. A clause that exits at once with amplitude -1 gives exactly the same gate as the rotation with feedback. To learn which body we have, we put a phase on the return wire before we close it. Only the rotation sends amplitude through that wire. We can compute the response from the table or probe it with access to the wire, while the exits still interfere. In `trace (guardRound; phaseBack)` the second iso passes `exit unit` straight through and multiplies `back unit` by $z = e^{i\varphi}$. For $z = -1$ its return clause reads `back unit` ↔ `-1 . back unit`.



The direct $3/5$ misses the gate, the first return picks up z and gives $-(16/25)z$, and the next gives $-(48/125)z^2$. With $t = 3/5$ and $s = 4/5$, each further trip contributes another factor tz, giving

$$\Theta_{\mathrm{rot}}(z) = t - s^2 z(1 + tz + t^2 z^2 + \cdots) = \frac{t - z}{1 - tz}.$$

On `exit unit` the first call sends amplitude $s = 4/5$ back, so a guard measurement gives `back` with probability $|s|^2 = 16/25$. The next call then starts on `back unit`, where its return amplitude is $t = 3/5$ and the probability of another return is $|t|^2 = 9/25$. The probabilities of making at least one, two and three returns are s^2, s^2t^2, s^2t^4. Summing these tails counts a run with k returns k times, so the mean is $s^2/(1 - t^2) = 1$ whenever $s \neq 0$. A smaller chance s^2 of entering is exactly offset by a longer stay, $1/s^2$ returns on average once inside.

When we close the wire coherently, the same returns contribute amplitudes $s, stz, s(tz)^2, \dots$ just before the phase gate. At $z = 1$ they add to $s/(1-t) = 2$ on unit input. The squared magnitude is the *circulating weight*, here $|2|^2 = 4$, four times the measured mean. At $z = -1$ the signs alternate, giving amplitude $s/(1+t) = 1/2$ and weight $1/4$.

To see this weight at the output, compare nearby phases. From the closed form $(t-z)/(1-tz)$,

$$\Theta_{\mathrm{rot}}(1) = \frac{t-1}{1-t} = -1, \qquad \frac{d\Theta_{\mathrm{rot}}}{dz} = \frac{t^2-1}{(1-tz)^2}, \qquad \Theta'_{\mathrm{rot}}(1) = -\frac{1+t}{1-t} = -\frac{8/5}{2/5} = -4.$$

Since $e^{i\delta} = 1 + i\delta + O(\delta^2)$, this gives $\Theta_{\mathrm{rot}}(e^{i\delta}) = -1 - 4i\delta + O(\delta^2)$, so its phase changes by 4δ, the weight times δ, to first order. Heebner and Boyd obtain the same factor for a lossless ring at resonance ($z = 1$): it is the intensity inside the ring divided by the incident intensity [29, p. 848, Eq. (7)]. If a trip adds phase $\omega\tau_0$ at angular frequency ω and the coupler is independent of frequency, the delay relative to the direct path is $4\tau_0$ [50, 56].

From one return wire to a finite body. Write $F_z u$ for the return leaving the body just before the phase gate. The first call puts Cu there. The gate feeds $zF_z u$ back in, so D contributes $zDF_z u$ to the next return and B contributes $zBF_z u$ to the external output. On the emitting space Y_1 the powers of $D_1 = D|_{Y_1}$ decay (Theorem 4.1), so $F_z = (I_{Y_1} - zD_1)^{-1}C$. Andrés-Martínez evaluates feedback frequency by frequency [3, Thm. 3.4.23]. The colligation identity gives the phase derivative [7, Eq. (4.10)], and monitored recurrence gives its average and dimension formula with the first call included [11, Eq. (8), Thm. 3.2].

Theorem 8.1 (The measured mean and the phase derivative). *For a finite unitary body, let x_k be the exit map after k returns. At $z = e^{i\varphi}$ the coherent output map is $\Theta(z) = A + zBF_z = \sum_{k\ge 0} z^k x_k$. Measuring the guard gives exit k with probability $\|x_k u\|^2$ on unit input u, so the mean number of returns is $\langle u, Nu\rangle$ for $N = \sum_{k\ge 0} k\, x_k^\dagger x_k$. Then*

$$Q(z) = -i\Theta(z)^\dagger \frac{d}{d\varphi}\Theta(e^{i\varphi}) = F_z^\dagger F_z, \qquad \frac{1}{2\pi}\int_0^{2\pi} Q(e^{i\varphi})\, d\varphi = N, \qquad \operatorname{tr} N = \dim Y_1.$$

At phases z and $w = ze^{i\delta}$ the body takes $(u, zF_z u)$ to $(\Theta(z)u, F_z u)$ and likewise at w. Preserving the inner product for this pair on unit input u gives

$$\langle \Theta(z)u, \Theta(w)u\rangle = 1 + (e^{i\delta} - 1)\langle F_z u, F_w u\rangle.$$

As w approaches z, the return vectors agree in the limit, so the imaginary part of the overlap is $\delta\|F_z u\|^2$ to first order. Its slope $\langle u, Q(z)u\rangle$ measures the circulating weight. For a fixed return count k, the output simply acquires z^k, so $\Theta(z) = z^k L$ and $Q(z) = kI$.

Averaging $\|F_z u\|^2$ over a full turn cancels the cross terms, leaving $\|Cu\|^2 + \|D_1 Cu\|^2 + \|D_1^2 Cu\|^2 + \cdots$. These are the probabilities of at least one return, at least two, and so on. Their sum counts a run with k returns k times. When the agreement test permits a measured replacement, this gives its expected cost in calls to the body: $1 + \langle u, Nu\rangle$, including the first call.

Summing these means over an orthonormal input basis gives $\operatorname{tr} N$. To see why this is a dimension, substitute $CC^\dagger = I_{Y_1} - D_1 D_1^\dagger$ into each tail. Writing $a_j = \operatorname{tr}(D_1^j D_1^{\dagger j})$, the jth tail contributes $a_j - a_{j+1}$. These differences sum to $a_0 = \dim Y_1$ since $D_1^j \to 0$ (Appendix E). For $X' = X$, counting the initial call gives the monitored sum $\dim X + \dim Y_1$ [11, Thm. 3.2, Eq. (9)].

Reading the weight with one qubit. We let a qubit turn a little each time amplitude crosses the return wire. The idea comes from the Larmor clock, which probes the time a particle spends in a region by letting a weak magnetic field turn its spin only there [13, 51].

In Figure 11 the probe starts in $(|0\rangle + |1\rangle)/\sqrt{2}$ and travels with the data, enlarging the return wire to $H_r \otimes Y$. On each return its branches pick up z and $ze^{i\delta}$, so after k returns their relative phase is $e^{ik\delta}$. After the exits have added, $S^\dagger$ multiplies the second branch by $-i$, so the final Hadamard reads the imaginary part of the overlap [15, p. 1, following Eq. (3)]:

$$\Pr(b=0) - \Pr(b=1) = \operatorname{Im}\langle \Theta(z)u, \Theta(ze^{i\delta})u\rangle = \delta\,\|F_z u\|^2 + O(\delta^2).$$

An immediate exit leaves the bits balanced, while `guardRound` at $z = 1$ gives a difference of $4\delta + O(\delta^2)$. Over n runs the frequency difference divided by δ estimates four with bias $O(\delta)$ and variance at most $1/(n\delta^2)$. Near $t = 1$ the peak narrows and the probe needs a smaller step (Appendix I).

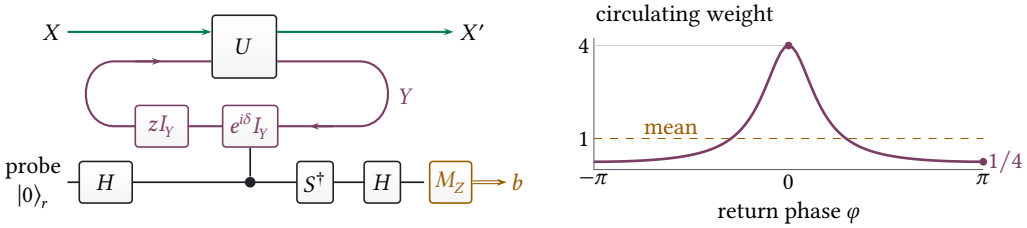


Fig. 11. Distributivity carries the probe through the external and returning ports. Before the final gates the state is $(|0\rangle \otimes \Theta(z)u + |1\rangle \otimes \Theta(ze^{i\delta})u)/\sqrt{2}$. The curve is the weight for `guardRound`, the dashed line its mean.

Averaging over the phase. Keep δ fixed and choose z uniformly for each run. Averaging z^{k-j} leaves only pairs with equal counts. Each surviving count k turns the probe by $k\delta$, so for `guardRound` the bit probabilities now differ by $\delta + O(\delta^2)$: the probe reads the measured mean of one. The same average on the data gives

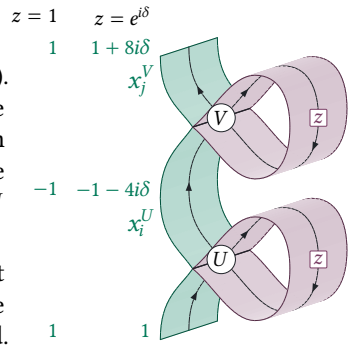
$$\mathcal{M}(\rho) = \frac{1}{2\pi}\int_0^{2\pi} \Theta(e^{i\varphi})\rho\Theta(e^{i\varphi})^\dagger\, d\varphi. \tag{8.1}$$

Without the probe the bypass caller reads 1 at $z = 1$ and 0 at $z = -1$. A fair coin choosing between them gives 1 half the time. Only odd counts change sign, so equal parities still interfere (Appendix C).

Over a full turn, the first Hadamard sends $1/\sqrt{2}$ down each path. The second subtracts the loop path from the bypass and divides by $\sqrt{2}$, giving amplitude $(1 - \Theta_{\text{rot}}(z))/2$ at output 1. Since $|\Theta_{\text{rot}}(z)| = 1$, its probability is $(1 - \operatorname{Re}\Theta_{\text{rot}}(z))/2$. Every delayed exit carries a power z^k whose average vanishes, leaving the immediate 3/5 and probability $(1 - 3/5)/2 = 1/5$.

Varying z gives `guardRound` an overall phase because every exit is a multiple of one gate. Conversely, a fixed data channel has the same value at $z = 1$ as its average, so coherent feedback and guard measurement agree (Appendix E).

Loops in sequence. At $z = 1$ two closed `guardRound` loops look like a straight wire because their signs cancel. We put the probe qubit on both return wires so that either loop can turn it by δ.



In `twoSections` we close both return ports together (Appendix F). Follow the external sheet from U to V. Each return picks up the same z on either wire, so i returns below and j above carry z^{i+j} with exit map $x_j^V x_i^U$. One trip below and two above thus carry the same z^3 as three below. Their amplitudes add, giving $x_k = \sum_{i+j=k} x_j^V x_i^U$ at total count k, and response $\Theta_{\text{rot}}(z)^2$ for two rotations.

In two identical rings a short pulse records the total. The first pulse passes straight through both rings. One trip later, the pulse from a trip in the first ring meets the pulse from a trip in the second.

At the frequency where a trip takes a whole number of cycles, the two sign changes cancel. But when we vary the common phase the responses multiply, so their phase shifts add. One ring multiplied the amplitude by $-1-4i\delta+O(\delta^2)$ at $z = e^{i\delta}$. The pair multiplies it by $(-1-4i\delta)^2 = 1+8i\delta+O(\delta^2)$, as labeled on the sheet, and the probe reads the imaginary part: a bit imbalance of $8\delta + O(\delta^2)$ for the pair and zero for a straight wire. At every z the weights add. Each averages over z to one, its measured mean (Theorem 8.1), so the pair averages two. The probe cannot tell which loop returned, so a return below and one above still interfere.

9 Further related work and future directions

Sabry et al. iterate by recurring on smaller arguments [48], whereas closing the return wire lets arbitrarily many trips contribute to a single clause. Clément and Perdrix eliminate feedback in polarizing beam-splitter diagrams on at least two wires with invertible gates, where a path uses each feedback wire at most twice [16, Prop. 26]. For our unitary tables Andrés-Martínez's construction sums all trips through each port [3, Thm. 3.3.15].

A fresh test qubit distinguishes calls three and five [4, Fig. 2 and § 4], whereas our parity bit gives them the same label. Andrés-Martínez varies how well each test distinguishes the guard values [3, § 4.2]. Silq checks when a temporary can be uncomputed [9]. For a count, our test tells us when discarding it already suffices. A runtime transformer computes the expected cost of measured iteration [43], while the probe recovers its mean return count from interfering exits.

Future work. Separate counters remember which loop returned. Whether a shared total suffices can be decided by comparing the data channels (Appendix D). We would like to infer this from the clauses, also for recursive calls that reach the same depth by different routes.

10 Conclusion

Can a quantum computer run a loop? For a finite reversible body we have three ways to do it. We can keep the count in a register that advances on each return, measure the guard after each call to decide whether to repeat, or leave the count implicit and compile coherent feedback into a gate. They agree on the data exactly when the first $m + 1$ exits are multiples of one exit (Theorem 7.1).

All three read the series $A + B(I + D + D^2 + \cdots)C$. Bartha's Kleene formula takes its limit, and the term BD^kC is our exit x_{k+1} [6, Eq. (1)]. Over the Booleans the same sum is reachability, $I \vee D \vee D^2 \vee \cdots$. Its terms can only add: a path after five returns cannot undo one after three, so forgetting the count loses nothing. Amplitudes, however, can cancel, as in $3/5 - 16/25 - \cdots = -1$. The coherent loop adds them, a counter keeps them apart, and measurement adds their probabilities.

The choice recurs wherever a program's history stays quantum. A set in superposition needs a history-independent representation: if two insertion orders leave $[1, 2]$ and $[2, 1]$ for the same set, their amplitudes no longer cancel [61, § 1.2, Def. 3.1]. Two loops in sequence show this. The first sends $|0\rangle$ straight through and $|1\rangle$ once around, the second the reverse, so on $|+\rangle$ both branches make one return. A shared counter reads 1 on both, like the set, leaving $|+\rangle$. A counter on each loop reads $(0, 1)$ beside $|0\rangle$ and $(1, 0)$ beside $|1\rangle$, like the two lists, and discarding them leaves an equal mixture (Appendix F). We knew all along that there was one return. What changed the qubit was writing down where it happened. Even a counter, the simplest data structure, has a history that a program can keep, measure or erase. Each has a consequence, and only erased histories interfere.

AI Use. The authors used ChatGPT Codex (GPT-6.1 Sol) to revise and format prose, generate TeX for figures, survey the literature (especially in optics), implement allocation and hiding, and check their reasoning, and Opus 5.5 for manuscript review. They verified the results and take responsibility for the paper.

Data-Availability Statement

The accompanying artifact contains the ISO interpreter, the paper's runnable examples, an exact checker for real rational clause tables, and a script for comparing the checker with a direct channel calculation. We plan to submit the artifact for evaluation if the paper is accepted.

References

[1] Samson Abramsky, Esfandiar Haghverdi, and Philip Scott. 2002. Geometry of Interaction and linear combinatory algebras. *Mathematical Structures in Computer Science* 12, 5 (2002), 625–665. doi:10.1017/S0960129502003730.

[2] Paul M. Alsing, Edwin E. Hach, III, Christopher C. Tison, and A. Matthew Smith. 2017. Quantum-optical description of losses in ring resonators based on field-operator transformations. *Physical Review A* 95 (2017), 053828. arXiv:1608.01280 [quant-ph]. doi:10.1103/PhysRevA.95.053828.

[3] Pablo Andrés-Martínez. 2022. *Unbounded loops in quantum programs: categories and weak while loops.* Ph. D. Dissertation. University of Edinburgh. arXiv:2212.05371. doi:10.48550/arXiv.2212.05371.

[4] Nicola Assolini and Alessandra Di Pierro. 2025. A Denotational Semantics for Quantum Loops. arXiv:2506.23320 [cs.PL]. doi:10.48550/arXiv.2506.23320.

[5] Costin Bădescu and Prakash Panangaden. 2015. Quantum Alternation: Prospects and Problems. In *Proceedings of the 12th International Workshop on Quantum Physics and Logic (QPL 2015) (EPTCS, Vol. 195).* Open Publishing Association, Waterloo, NSW, Australia, 33–42. doi:10.4204/EPTCS.195.3.

[6] Miklós Bartha. 2014. Quantum Turing automata. In *Proceedings of DCM 2012 (EPTCS, Vol. 143).* Open Publishing Association, Waterloo, NSW, Australia, 17–31. arXiv:1404.0074. doi:10.4204/EPTCS.143.2.

[7] Jussi Behrndt, Seppo Hassi, and Henk de Snoo. 2009. Boundary Relations, Unitary Colligations, and Functional Models. *Complex Analysis and Operator Theory* 3, 1 (2009), 57–98. doi:10.1007/s11785-008-0064-z.

[8] Henri Bergson. 1910. *Time and Free Will: An Essay on the Immediate Data of Consciousness.* Translated by F. L. Pogson. Swan Sonnenschein, London.

[9] Benjamin Bichsel, Maximilian Baader, Timon Gehr, and Martin Vechev. 2020. Silq: A high-level quantum language with safe uncomputation and intuitive semantics. In *PLDI*, 286–300. doi:10.1145/3385412.3386007.

[10] Niels Bohr. 1949. Discussion with Einstein on epistemological problems in atomic physics. In *Albert Einstein: Philosopher-Scientist*, Paul Arthur Schilpp (Ed.). The Library of Living Philosophers, Evanston, IL, 199–241.

[11] J. Bourgain, F. A. Grünbaum, L. Velázquez, and J. Wilkening. 2014. Quantum Recurrence of a Subspace and Operator-Valued Schur Functions. *Communications in Mathematical Physics* 329, 3 (2014), 1031–1067. arXiv:1302.7286. doi:10.1007/s00220-014-1929-9.

[12] M. S. Brodskii. 1978. Unitary operator colligations and their characteristic functions. *Russian Mathematical Surveys* 33, 4 (1978), 159–191. doi:10.1070/RM1978v033n04ABEH002495.

[13] M. Büttiker. 1983. Larmor precession and the traversal time for tunneling. *Physical Review B* 27, 10 (1983), 6178–6188. doi:10.1103/PhysRevB.27.6178.

[14] Jacques Carette, Chris Heunen, Robin Kaarsgaard, and Amr Sabry. 2024. How to Bake a Quantum Π. *Proc. ACM Program. Lang.* 8, ICFP, Article 236 (2024), 29 pages. doi:10.1145/3674625.

[15] Andrew M. Childs. 2013. Quantum Algorithms, Lecture 18: Approximating the Jones Polynomial. University of Waterloo lecture notes. https://www.cs.umd.edu/~amchilds/teaching/w13/l18.pdf.

[16] Alexandre Clément and Simon Perdrix. 2020. PBS-Calculus: A Graphical Language for Coherent Control of Quantum Computations. In *45th International Symposium on Mathematical Foundations of Computer Science (LIPIcs, Vol. 170).* Schloss Dagstuhl, Dagstuhl, Germany, 24:1–24:14. doi:10.4230/LIPIcs.MFCS.2020.24.

[17] Cole Comfort, Antonin Delpeuch, and Jules Hedges. 2020. Sheet diagrams for bimonoidal categories. arXiv:2010.13361. doi:10.48550/arXiv.2010.13361.

[18] Andrew W. Cross, Ali Javadi-Abhari, Thomas Alexander, Niel de Beaudrap, Lev S. Bishop, Steven Heidel, Colm A. Ryan, Prasahnt Sivarajah, John Smolin, Jay M. Gambetta, and Blake R. Johnson. 2022. OpenQASM 3: A Broader and Deeper Quantum Assembly Language. *ACM Trans. Quantum Comput.* 3, 3, Article 12. doi:10.1145/3505636.

[19] Marek Czachor. 2019. Time travel without paradoxes: Ring resonator as a universal paradigm for looped quantum evolutions. *Physics Letters A* 383 (2019), 2704–2712. arXiv:1805.12129v8. doi:10.1016/j.physleta.2019.05.043.

[20] Aaron David Fairbanks and Peter Selinger. 2025. On traces in categories of contractions. In *Proceedings of the 22nd International Conference on Quantum Physics and Logic (QPL 2025) (EPTCS, Vol. 426)*, Alejandro Díaz-Caro, Ognyan Oreshkov, and Ana Belén Sainz (Eds.). Open Publishing Association, Waterloo, NSW, Australia, 79–99. arXiv:2502.14993. doi:10.4204/EPTCS.426.3. Journal version: *Math. Struct. Comput. Sci.* 36 (2026), e4. doi:10.1017/S0960129525100431.

[21] Y. V. Fyodorov and H.-J. Sommers. 1997. Statistics of resonance poles, phase shifts and time delays in quantum chaotic scattering: Random matrix approach for systems with broken time-reversal invariance. *Journal of Mathematical Physics* 38 (1997), 1918–1981. doi:10.1063/1.531919.

[22] Jean-Yves Girard. 1995. Geometry of interaction III: accommodating the additives. In *Advances in Linear Logic*. Cambridge University Press, 329–389. doi:10.1017/CBO9780511629150.017.

[23] Robert Glück and Robin Kaarsgaard. 2018. A Categorical Foundation for Structured Reversible Flowchart Languages: Soundness and Adequacy. *Logical Methods in Computer Science* 14, 3, Article 16 (2018), 38 pages. arXiv:1710.03666. doi:10.23638/LMCS-14(3:16)2018.

[24] John E. Gough. 2014. Feedback network models for quantum transport. *Physical Review E* 90 (2014), Article 062109. doi:10.1103/PhysRevE.90.062109.

[25] John Gough and Matthew R. James. 2009. Quantum Feedback Networks: Hamiltonian Formulation. *Communications in Mathematical Physics* 287 (2009), 1109–1132. doi:10.1007/s00220-008-0698-8.

[26] F. A. Grünbaum, L. Velázquez, A. H. Werner, and R. F. Werner. 2013. Recurrence for Discrete Time Unitary Evolutions. *Communications in Mathematical Physics* 320 (2013), 543–569. doi:10.1007/s00220-012-1645-2.

[27] Masahito Hasegawa. 1997. Recursion from Cyclic Sharing: Traced Monoidal Categories and Models of Cyclic Lambda Calculi. In *Typed Lambda Calculi and Applications (Lecture Notes in Computer Science, Vol. 1210)*. Springer, 196–213. doi:10.1007/3-540-62688-3_37.

[28] S. W. Hawking. 1992. Chronology protection conjecture. *Physical Review D* 46, 2 (1992), 603–611. doi:10.1103/PhysRevD.46.603.

[29] John E. Heebner and Robert W. Boyd. 1999. Enhanced all-optical switching by use of a nonlinear fiber ring resonator. *Optics Letters* 24, 12 (1999), 847–849. doi:10.1364/OL.24.000847.

[30] Chris Heunen and Robin Kaarsgaard. 2022. Quantum information effects. *Proc. ACM Program. Lang.* 6, POPL, Article 2 (2022), 27 pages. arXiv:2107.12144. doi:10.1145/3498663.

[31] Mathieu Huot and Sam Staton. 2019. Universal Properties in Quantum Theory. In *Proceedings of the 15th International Conference on Quantum Physics and Logic (QPL 2018) (EPTCS, Vol. 287)*. Open Publishing Association, Waterloo, NSW, Australia, 213–223. doi:10.4204/EPTCS.287.12.

[32] Mathieu Huot and Sam Staton. 2019. Quantum Channels as a Categorical Completion. In *34th Annual ACM/IEEE Symposium on Logic in Computer Science (LICS)* (Vancouver, BC, Canada). IEEE, 1–13. doi:10.1109/LICS.2019.8785700.

[33] Roshan P. James and Amr Sabry. 2012. Information Effects. In *Proceedings of the 39th ACM SIGPLAN-SIGACT Symposium on Principles of Programming Languages (POPL '12)*. ACM, New York, NY, USA, 73–84. doi:10.1145/2103656.2103667.

[34] Roshan P. James and Amr Sabry. 2013. Isomorphic Interpreters from Logically Reversible Abstract Machines. In *Reversible Computation: 4th International Workshop, RC 2012, Revised Papers (Lecture Notes in Computer Science, Vol. 7581)*. Springer, Berlin, Heidelberg, 57–71. doi:10.1007/978-3-642-36315-3_5.

[35] André Joyal, Ross Street, and Dominic Verity. 1996. Traced monoidal categories. *Mathematical Proceedings of the Cambridge Philosophical Society* 119, 3 (1996), 447–468. doi:10.1017/S0305004100074338.

[36] Robin Kaarsgaard and Mathys Rennela. 2021. Join Inverse Rig Categories for Reversible Functional Programming, and Beyond. In *Proceedings of the 37th Conference on the Mathematical Foundations of Programming Semantics (MFPS 2021) (EPTCS, Vol. 351)*. Open Publishing Association, Waterloo, NSW, Australia, 152–167. arXiv:2105.09929. doi:10.4204/EPTCS.351.10.

[37] Louis Lemonnier. 2024. *The Semantics of Effects: Centrality, Quantum Control and Reversible Recursion*. Ph. D. Dissertation. Université Paris-Saclay. arXiv:2406.07216. doi:10.48550/arXiv.2406.07216.

[38] Noah Linden and Sandu Popescu. 1998. The halting problem for quantum computers. arXiv:quant-ph/9806054. doi:10.48550/arXiv.quant-ph/9806054.

[39] Octavio Malherbe, Philip Scott, and Peter Selinger. 2012. Partially traced categories. *Journal of Pure and Applied Algebra* 216, 12 (2012), 2563–2585. arXiv:1107.3608. doi:10.1016/j.jpaa.2012.03.026.

[40] Samuel J. Mason. 1953. Feedback Theory: Some Properties of Signal Flow Graphs. Technical Report 153. Research Laboratory of Electronics, MIT.

[41] Microsoft. 2025. Conditional loops in Q#. *Microsoft Learn*. https://learn.microsoft.com/en-us/azure/quantum/user-guide/language/expressions/conditionalloops.

[42] John M. Myers. 1997. Can a Universal Quantum Computer Be Fully Quantum? *Physical Review Letters* 78 (1997), 1823–1824. doi:10.1103/PhysRevLett.78.1823.

[43] Federico Olmedo and Alejandro Díaz-Caro. 2019. Runtime Analysis of Quantum Programs: A Formal Approach. arXiv:1911.11247.

[44] OpenQASM Contributors. [n. d.]. Classical instructions. *OpenQASM Live Specification*, section on while loops. Accessed November 8, 2026. https://openqasm.com/language/classical.html#while-loops.

[45] Adam Paetznick and Krysta M. Svore. 2014. Repeat-Until-Success: Non-deterministic decomposition of single-qubit unitaries. *Quantum Information and Computation* 14, 15–16 (2014), 1277–1301. doi:10.26421/QIC14.15-16-2.

[46] M. G. Raymer and C. J. McKinstrie. 2013. Quantum Input-Output Theory for Optical Cavities with Arbitrary Coupling Strength: Application to Two-Photon Wave-Packet Shaping. *Physical Review A* 88 (2013), 043819. arXiv:1309.5915 [quant-ph]. doi:10.1103/PhysRevA.88.043819.

[47] Michael Reck, Anton Zeilinger, Herbert J. Bernstein, and Philip Bertani. 1994. Experimental realization of any discrete unitary operator. *Physical Review Letters* 73 (1994), 58–61. doi:10.1103/PhysRevLett.73.58.

[48] Amr Sabry, Benoît Valiron, and Juliana Kaizer Vizzotto. 2018. From symmetric pattern-matching to quantum control. In *FoSSaCS 2018*. Springer, Cham, 348–364. arXiv:1804.00952. doi:10.1007/978-3-319-89366-2_19.

[49] Peter Selinger. 2004. Towards a quantum programming language. *Mathematical Structures in Computer Science* 14, 4 (2004), 527–586. doi:10.1017/S0960129504004256.

[50] Felix T. Smith. 1960. Lifetime Matrix in Collision Theory. *Physical Review* 118, 1 (1960), 349–356. doi:10.1103/PhysRev.118.349.

[51] Aephraim M. Steinberg. 1995. How much time does a tunneling particle spend in the barrier region? *Physical Review Letters* 74 (1995), 2405–2409. doi:10.1103/PhysRevLett.74.2405.

[52] Nikolas Tezak, Armand Niederberger, Dmitri S. Pavlichin, Gopal Sarma, and Hideo Mabuchi. 2012. Specification of photonic circuits using Quantum Hardware Description Language. *Philosophical Transactions of the Royal Society A* 370 (2012), 5270–5290. doi:10.1098/rsta.2011.0526.

[53] Matt Visser. 1993. From wormhole to time machine: Remarks on Hawking's chronology protection conjecture. *Physical Review D* 47, 2 (1993), 554–565. doi:10.1103/PhysRevD.47.554.

[54] Songli Wang, Yuyang Wang, Swarnava Sanyal, Robert Parsons, Xingchen Ji, Yoshitomo Okawachi, Xiang Meng, Michal Lipson, Alexander Gaeta, and Keren Bergman. 2025. Automated tuning of a ring-assisted MZI-based interleaver for Kerr frequency combs. *Optics Letters* 50, 2 (2025), 698–701. doi:10.1364/OL.546772.

[55] John Watrous. 2018. *The Theory of Quantum Information*. Cambridge University Press, Cambridge, UK. doi:10.1017/9781316848142.

[56] Eugene P. Wigner. 1955. Lower Limit for the Energy Derivative of the Scattering Phase Shift. *Physical Review* 98 (1955), 145–147. doi:10.1103/PhysRev.98.145.

[57] Mingsheng Ying. 2011. Floyd–Hoare Logic for Quantum Programs. *ACM Transactions on Programming Languages and Systems* 33, 6, Article 19 (2011), 49 pages. arXiv:0906.4586v1. doi:10.1145/2049706.2049708.

[58] Mingsheng Ying. 2014. Quantum Recursion and Second Quantisation. arXiv:1405.4443 [quant-ph]. doi:10.48550/arXiv.1405.4443.

[59] Mingsheng Ying and Yuan Feng. 2010. Quantum loop programs. *Acta Informatica* 47, 4 (2010), 221–250. arXiv:quant-ph/0605218. doi:10.1007/s00236-010-0117-4.

[60] Mingsheng Ying, Nengkun Yu, and Yuan Feng. 2012. Defining Quantum Control Flow. arXiv preprint. arXiv:1209.4379v1.

[61] Charles Yuan and Michael Carbin. 2022. Tower: data structures in quantum superposition. *Proc. ACM Program. Lang.* 6, OOPSLA2, Article 134 (2022), 41 pages. arXiv:2205.10255. doi:10.1145/3563297.

[62] Charles Yuan, Agnes Villanyi, and Michael Carbin. 2024. Quantum control machine: the limits of control flow in quantum programming. *Proc. ACM Program. Lang.* 8, OOPSLA1, Article 94 (2024), 28 pages. arXiv:2304.15000. doi:10.1145/3649811.

[63] Zhicheng Zhang and Mingsheng Ying. 2025. Quantum register machine: efficient implementation of quantum recursive programs. *Proc. ACM Program. Lang.* 9, PLDI, Article 180 (2025), 26 pages. arXiv:2408.10054. doi:10.1145/3729283.

Citations in these appendices refer to the reference list of the paper.

A Typing and interpreting the clauses

A.1 Syntax

In `trace guardRound on unit`, `trace guardRound` closes the return wire of the named table and `on` applies the resulting iso to `unit`. Figure 12 gives the grammar. Types are built from 0 and 1 by sum and tensor, written `+` and `x` in source types. In programs, `;` denotes sequence, `+` direct sum, `*` tensor and `dagger` the adjoint. Before matching or typing, recursively replace exit v and back v

$$
\begin{aligned}
a &::= 0 \mid 1 \mid a_1 + a_2 \mid a_1 \otimes a_2 \\
v &::= * \mid x \mid \mathsf{left}\ v \mid \mathsf{right}\ v \mid \langle v_1, v_2 \rangle \mid \mathsf{exit}\ v \mid \mathsf{back}\ v \\
p, q &::= * \mid x \mid \langle p, q \rangle \\
e &::= v \mid \textstyle\sum_i \alpha_i e_i \mid \mathsf{let}\ p = \omega\, q\ \mathsf{in}\ e \\
\omega &::= f \mid \{v_i \leftrightarrow e_i\}_{i \in I} \mid \omega\,;\,\omega \mid \omega \oplus \omega \mid \omega \otimes \omega \mid \omega^{\dagger} \mid \mathsf{trace}_c\ \omega \\
t &::= e \mid \omega\, t \\
\mathit{decl} &::= f :: a \leftrightarrow b\,;\ f = \omega \mid \mathsf{round}\ f\ :\ a + c = \{v_i \leftrightarrow e_i\}_{i \in I}
\end{aligned}
$$

Fig. 12. Finite iso definitions. A name f refers to an earlier closed definition. The coefficient α_i is a complex scalar. A signature uses `::`, while the `round` abbreviation uses `:`. In trace_c, the return has type c.

by left v and right v, preserving their payloads. Section 3 gives the other surface abbreviations. Listings render `<->` as $\leftrightarrow$ and `<, >` as $\langle, \rangle$. The surface prefix operators (including `dagger`, written $\omega^{\dagger}$ in formulas) bind more tightly than `*`, then `+`, then `;`. The outer `on` applies the entire program to its input, written $\omega\, t$ in the core grammar.

A.2 Statics

Checking and evaluation use the pair $(|\Sigma|, s)$ in lexicographic order, where s is the size of the subprogram. Unfolding a name shortens the declaration prefix, while checking a clause or its local call descends into a proper subterm. At each such call we use the checked matrix of the smaller iso. Matrix combinators, once elaborated, evaluate by structural recursion.

Write Σ ok for a checked prefix. Both declaration forms add a fresh name whose body checks in that prefix. Write $\Sigma \vdash d \Rightarrow \Sigma'$ for the extension and $\sigma = (a + c) \leftrightarrow (a + c)$ for the type of a `round`.

$$
\frac{}{\emptyset\ \mathsf{ok}} \qquad \frac{\Sigma\ \mathsf{ok} \quad f \notin \mathrm{dom}(\Sigma) \quad \Sigma \vdash \omega\ :\ a \leftrightarrow b}{\Sigma, (f = \omega\ :\ a \leftrightarrow b)\ \mathsf{ok}}.
$$

An ordinary declaration adds $(f = \omega\ :\ a \leftrightarrow b)$. Under the same freshness premise, $\Sigma \vdash \mathscr{T}\ :\ \sigma$ gives $\Sigma \vdash \mathtt{round}\ f\ :\ a + c = \mathscr{T} \Rightarrow \Sigma, (f = \mathscr{T}\ :\ \sigma)$.

Patterns and expressions. A context Δ gives a type to each pattern variable. We write Δ_1, Δ_2 when the two contexts have disjoint domains. The values we may assign to a variable of type a come from its finite basis:

$$
\mathscr{B}(0) = \emptyset, \qquad \mathscr{B}(1) = \{*\},
$$
$$
\mathscr{B}(a + b) = \mathsf{left}\,\mathscr{B}(a)\ \sqcup\ \mathsf{right}\,\mathscr{B}(b), \quad \mathscr{B}(a \otimes b) = \mathscr{B}(a) \times \mathscr{B}(b).
$$

Since $\mathscr{B}(0)$ is empty, the empty table gives the unique map on 0. Whenever we choose a unit input or require the exits to span a line, we assume the external input type has at least one basis value.

The pattern judgment $\Delta \vdash v : a$ binds one basis value per variable:

$$\emptyset \vdash * : 1 \qquad x : a \vdash x : a \qquad \frac{\Delta \vdash v : a}{\Delta \vdash \mathrm{left}\ v : a + b} \qquad \frac{\Delta \vdash v : b}{\Delta \vdash \mathrm{right}\ v : a + b}$$

$$\frac{\Delta_1 \vdash v : a \qquad \Delta_2 \vdash w : b}{\Delta_1, \Delta_2 \vdash \langle v, w \rangle : a \otimes b}.$$

An assignment $\gamma \in \mathscr{B}(\Delta) = \prod_{(x:a)\in\Delta} \mathscr{B}(a)$ chooses one basis value for each variable. There is one way to assign no variables and none to assign a variable of empty type. Write $\mathscr{H}(\Delta) = \mathbb{C}^{\mathscr{B}(\Delta)}$ for the space with these assignments as its basis.

Substitution builds $v[\gamma]$ from the assignment, and matching recovers γ from that value. For unit there is nothing to bind, a variable binds the value itself, and a pair matches its two components in disjoint contexts. A tag restricts which values match, but on a matching value we remove the tag and continue. Without tag tests, a product pattern therefore gives a bijection onto the whole basis, including when the basis is empty.

This is why a local call can use q to assemble its argument and p to bind every possible result. A variable in either position may carry a whole tagged value. The expression rules use these bindings in the judgment $\Sigma; \Delta \vdash e : a$.

$$\frac{\Delta \vdash v : a}{\Sigma; \Delta \vdash v : a} \qquad \frac{\Sigma; \Delta \vdash e_i : a\ (i \in I) \qquad \alpha_i \in \mathbb{C}}{\Sigma; \Delta \vdash \sum_{i\in I} \alpha_i e_i : a}$$

$$\frac{\Delta_1 \vdash q : c \qquad \Sigma \vdash \omega : c \leftrightarrow b \qquad \Delta_p \vdash p : b \qquad \Sigma; \Delta_2, \Delta_p \vdash e : a}{\Sigma; \Delta_1, \Delta_2 \vdash \mathsf{let}\ p = \omega\, q\ \mathsf{in}\ e : a}.$$

The result pattern is fresh and irrefutable by the bijection above. An empty sum in any context denotes the zero map from $\mathscr{H}(\Delta)$. For instance, left x in $0 + 1 \leftrightarrow 1$ may have this right-hand side under $x : 0$ because it contributes no column. A zero column on an inhabited input would violate the Gram test. A closed term is a closed expression or an application to a closed term.

$$\frac{\Sigma; \emptyset \vdash e : a}{\Sigma \vdash e : a} \qquad \frac{\Sigma \vdash t : a \qquad \Sigma \vdash \omega : a \leftrightarrow b}{\Sigma \vdash \omega\, t : b}.$$

The matching clause gives column s of M on input s. Its Gram entry $(M^\dagger M)_{ss'} = \sum_t \overline{M_{ts}} M_{ts'}$ tests the squared norm when $s = s'$ and the inner product otherwise. Once the columns are known, the dense test takes $O(d^3)$ operations for d basis inputs. Figure 13 shows the test on `guardRound`.

$$\frac{\begin{array}{c}\Delta_i \vdash v_i : a \quad \Sigma; \Delta_i \vdash e_i : b \quad (i \in I) \qquad \big(v_i[\mathscr{B}(\Delta_i)]\big)_{i\in I} \text{ disjointly covers } \mathscr{B}(a) \\ \Sigma; \gamma \vdash e_i \Downarrow \sum_{t\in\mathscr{B}(b)} M_{t,v_i[\gamma]} |t\rangle \quad (i \in I,\ \gamma \in \mathscr{B}(\Delta_i)) \qquad |\mathscr{B}(a)| = |\mathscr{B}(b)| \quad M^\dagger M = I\end{array}}{\Sigma \vdash \{v_i \leftrightarrow e_i\}_{i\in I} : a \leftrightarrow b}$$

The ground instances $v_i[\mathscr{B}(\Delta_i)]$ partition the input basis; empty parts contribute no columns. Over an exact coefficient domain such as $\mathbb{Q}(i, \sqrt{2})$ the rule is decidable. For unrestricted mathematical complex parameters it defines a predicate.

Reversible combinators. For $M \in \mathbb{C}^{\mathscr{B}(b)\times\mathscr{B}(a)}$ the atom rule is

$$\frac{|\mathscr{B}(a)| = |\mathscr{B}(b)| \qquad M^\dagger M = I}{[M] : a \leftrightarrow b}.$$

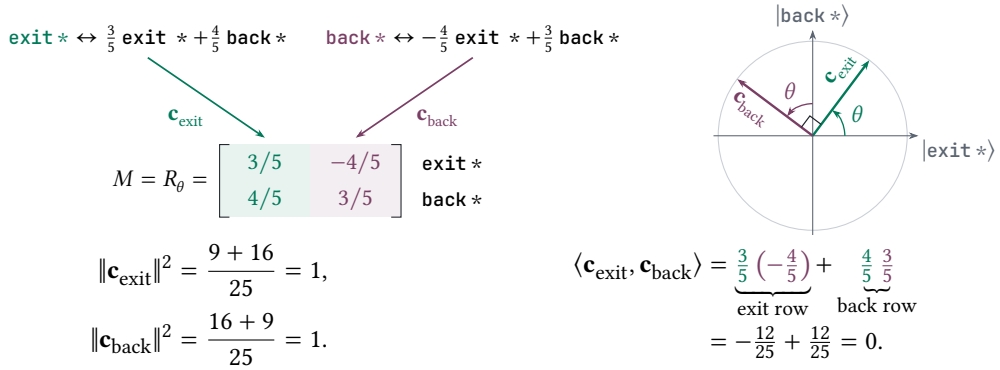


Fig. 13. Each clause supplies one column of R_θ. Here $*$ is `unit`. Both basis vectors rotate through θ and stay perpendicular. The products pair entries from the same row, conjugating the first for complex columns.

Write ω, ν for reversible programs and omit the common Σ in the following rules. A bound name has the type at its declaration. For finite types we have the structural isos of Π as abbreviations for the corresponding permutation clauses, including identity, symmetries and distributivity [34, § 2]. Here I_a fixes each basis value, $\sigma_{a,b} : a \otimes b \leftrightarrow b \otimes a$ exchanges the components, and $\mathrm{unit}_\otimes : a \otimes 1 \leftrightarrow a$ drops the unique $*$. Distributivity sends $(j, \mathrm{in}_i\, x)$ to $\mathrm{in}_i(j, x)$, with $j : [K]$ in d_K of Appendix C. The reversible rules are

$$\frac{f : a \leftrightarrow b \in \Sigma}{f : a \leftrightarrow b} \qquad \frac{\omega : a \leftrightarrow b \qquad \nu : b \leftrightarrow d}{\omega\,;\nu : a \leftrightarrow d} \qquad \frac{\omega : a \leftrightarrow b}{\omega^\dagger : b \leftrightarrow a}$$

$$\frac{\omega : a \leftrightarrow b \qquad \nu : c \leftrightarrow d}{\omega \oplus \nu : a + c \leftrightarrow b + d} \qquad \frac{\omega : a \leftrightarrow b \qquad \nu : c \leftrightarrow d}{\omega \otimes \nu : a \otimes c \leftrightarrow b \otimes d} \qquad \frac{\beta : a + c \leftrightarrow b + c}{\mathrm{trace}_c\, \beta : a \leftrightarrow b}.$$

A.3 Dynamics

Evaluating an expression. Let $\gamma \in \mathscr{B}(\Delta)$ assign basis values to the variables in a linear context Δ. The judgment $\Sigma; \gamma \vdash e \Downarrow \psi$ evaluates a checked expression to a finite output vector. Values instantiate their bindings and sums use the same assignment in every branch.

$$\frac{}{\Sigma; \gamma \vdash v \Downarrow |v[\gamma]\rangle} \qquad \frac{\Sigma; \gamma \vdash e_i \Downarrow \psi_i \ (i \in I)}{\Sigma; \gamma \vdash \sum_i \alpha_i e_i \Downarrow \sum_i \alpha_i \psi_i}.$$

For a local call, the typing rule splits the input into the argument bindings γ_1 and the bindings γ_2 retained by the continuation. For each basis output t of the call, the result pattern determines γ_t uniquely.

$$\frac{\Sigma \vdash \omega \Downarrow U \qquad a_t = U_{t,q[\gamma_1]} \qquad p[\gamma_t] = t \qquad \Sigma; \gamma_2 \uplus \gamma_t \vdash e \Downarrow \psi_t \qquad (t \in \mathscr{B}(b))}{\Sigma; \gamma_1 \uplus \gamma_2 \vdash \mathsf{let}\ p = \omega q\ \mathsf{in}\ e \Downarrow \sum_t a_t \psi_t}.$$

Closed expressions use the empty assignment, and closed applications apply the matrix of the checked iso.

$$\frac{\Sigma; \varnothing \vdash e \Downarrow \psi}{\Sigma \vdash e \Downarrow \psi} \qquad \frac{\Sigma \vdash \omega \Downarrow U \qquad \Sigma \vdash t \Downarrow \psi}{\Sigma \vdash \omega t \Downarrow U\psi}.$$

Write $E_e(s; \gamma)$ for the amplitude of output s when we evaluate e in environment γ. In a local application this gives

$$E_{\mathrm{let}}(s;\gamma_1,\gamma_2) = \sum_{t\in\mathscr{B}(b)} U_{t,q[\gamma_1]}\, E_e(s;\gamma_2,\gamma_t).$$

The sum multiplies the continuation columns by the selected amplitudes of U, carrying γ_2 through without measurement or normalization.

Elaboration and matrix evaluation. The matrix programs have grammar

$$\kappa ::= [M] \mid \kappa;\kappa \mid \kappa \oplus \kappa \mid \kappa \otimes \kappa \mid \kappa^\dagger \mid \mathrm{trace}_c\, \kappa.$$

For a checked table $\mathscr{T}$, let M be its matrix. For a named body ω_f, let Σ_f be its checked declaration prefix. Elaboration then acts as follows.

$$\begin{aligned}
\mathscr{E}_\Sigma(\mathscr{T}) &= [M], & \mathscr{E}_\Sigma(f) &= \mathscr{E}_{\Sigma_f}(\omega_f),\\
\mathscr{E}_\Sigma(\omega;\nu) &= \mathscr{E}_\Sigma(\omega);\mathscr{E}_\Sigma(\nu), & \mathscr{E}_\Sigma(\omega\oplus\nu) &= \mathscr{E}_\Sigma(\omega)\oplus\mathscr{E}_\Sigma(\nu),\\
\mathscr{E}_\Sigma(\omega\otimes\nu) &= \mathscr{E}_\Sigma(\omega)\otimes\mathscr{E}_\Sigma(\nu), & \mathscr{E}_\Sigma(\omega^\dagger) &= \mathscr{E}_\Sigma(\omega)^\dagger,\\
\mathscr{E}_\Sigma(\mathrm{trace}_c\,\omega) &= \mathrm{trace}_c\,\mathscr{E}_\Sigma(\omega).
\end{aligned}$$

For example, `trace guardRound` elaborates to $\mathrm{trace}_1[R_\theta]$. For $\kappa \Downarrow M$ and $\lambda \Downarrow N$, evaluation of the typed combinators gives

$$\begin{gathered}
[M] \Downarrow M, \qquad (\kappa;\lambda) \Downarrow NM,\\
(\kappa\oplus\lambda) \Downarrow M\oplus N, \quad (\kappa\otimes\lambda) \Downarrow M\otimes N, \quad \kappa^\dagger \Downarrow M^\dagger.
\end{gathered}$$

The feedback rule completes them:

$$\frac{\kappa \Downarrow U \quad U = \left(\begin{smallmatrix} A & B \\ C & D \end{smallmatrix}\right) \quad F : X \to Y \quad (I_Y - D)F = C}{\mathrm{trace}_c\,\kappa \Downarrow A + BF}.$$

Theorem 4.1 supplies a solution for the checked unitary body. All solutions give the same $A + BF$ even when $I - D$ is singular, and Proposition 4.2 computes it. Write $\Sigma \vdash \omega \Downarrow M$ when its elaboration evaluates to M.

A.4 Denotation

A table denotes the linear extension of its columns. Elaboration preserves these columns by the value, sum and local-call calculations above. The name case uses lookup. The combinator rules apply sequential matrices in execution order and take the direct sum, tensor or adjoint as indicated. Trace agrees with the exit sum (4.2) by Theorem 4.1. Equality on basis columns extends by linearity to inputs entangled with a retained factor.

Control of an endo-iso u is $I \oplus u$ under the distributive iso for $(1 + 1) \otimes a$. Replacing u by $e^{i\varphi}u$ leaves its channel unchanged but gives just the second branch a phase. The channel alone therefore does not determine the controlled gate, as in quantum alternation [5, § 2].

A.5 Effect extensions

Syntax. Write ω for a reversible iso, ν for an isometry program and h for a channel program. Their top-level applications form a separate syntactic class.

$$\begin{aligned}
\nu &::= \mathrm{arr}\,\omega \mid \mathsf{lift}\,\omega \mid \nu\,;\,\nu \mid \nu\oplus\nu \mid \nu\otimes\nu \mid \mathsf{count}_K\,\omega,\\
h &::= \mathrm{arr}\,\nu \mid \mathsf{hide}\,\nu \mid h\,;\,h \mid h\otimes h \mid \mathsf{while}_c\,\omega,\\
p_{\mathrm{top}} &::= \nu\ \mathsf{on}\ t \mid h\ \mathsf{on}\ t.
\end{aligned}$$

Statics. Isometries compose in sequence, by tensor and by direct sum.

$$\frac{v : a \rightarrowtail b \qquad w : b \rightarrowtail d}{v\,;w : a \rightarrowtail d} \qquad \frac{v : a \rightarrowtail b \qquad w : c \rightarrowtail d}{v \otimes w : a \otimes c \rightarrowtail b \otimes d} \qquad \frac{v : a \rightarrowtail b \qquad w : c \rightarrowtail d}{v \oplus w : a + c \rightarrowtail b + d}.$$

Channels also compose in sequence and by tensor product.

$$\frac{h : a \nrightarrow b \qquad k : b \nrightarrow d}{h\,;k : a \nrightarrow d} \qquad \frac{h : a \nrightarrow b \qquad k : c \nrightarrow d}{h \otimes k : a \otimes c \nrightarrow b \otimes d}.$$

Composition is in execution order and each layer embeds its identity iso. Allocation, hiding and the two loop constructs have the following rules.

$$\frac{\omega : a \leftrightarrow b}{\mathsf{arr}\,\omega : a \rightarrowtail b} \qquad \frac{\omega : a + d \leftrightarrow b}{\mathsf{lift}\,\omega : a \rightarrowtail b} \qquad \frac{\beta : a + c \leftrightarrow b + c \qquad K \geq 1}{\mathsf{count}_K\,\beta : a \rightarrowtail [K] \otimes b}$$

$$\frac{v : a \rightarrowtail b}{\mathsf{arr}\,v : a \nrightarrow b} \qquad \frac{v : a \rightarrowtail b \otimes g \qquad \mathscr{B}(g) \neq \emptyset}{\mathsf{hide}\,v : a \nrightarrow b} \qquad \frac{\beta : a + c \leftrightarrow b + c}{\mathsf{while}_c\,\beta : a \nrightarrow b}.$$

In a mixed sequence we read an iso at the isometry layer as $\mathsf{arr}\,\omega$ and at the channel layer as $\mathsf{arr}(\mathsf{arr}\,\omega)$. Since while requires a reversible body, a compiled inner trace can supply it, whereas an inner while already has channel type.

Once the program is built, we can apply it to a closed term t. An isometry returns a vector, and a channel returns an operator:

$$\frac{\Sigma \vdash t : a \qquad v : a \rightarrowtail b}{v\ \mathsf{on}\ t\ :\ \mathrm{Vec}(b)} \qquad \frac{\Sigma \vdash t : a \qquad h : a \nrightarrow b}{h\ \mathsf{on}\ t\ :\ \mathrm{Op}(b)}.$$

Denotation. Here $\mathrm{Vec}(b)$ consists of the vectors in $[\![b]\!]$ and $\mathrm{Op}(b)$ of the operators on that space. For $\Sigma \vdash t \Downarrow \psi$, the two applications are $v\ \mathsf{on}\ t \Downarrow [\![v]\!]\,\psi$ and $h\ \mathsf{on}\ t \Downarrow [\![h]\!](|\psi\rangle\langle\psi|)$. When ψ has unit norm, the channel returns a density matrix. The executable checks this input condition to numerical tolerance.

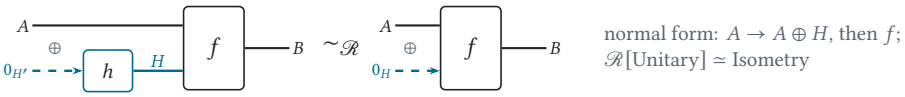
Figure 14 shows which parts of these maps the caller sees. An $\mathscr{R}[\mathsf{Unitary}]$ representative $U : X \oplus H \to X'$ acts on u by $U(u, 0)$, so changing its action on the unused summand leaves the exposed map unchanged. An $\mathscr{L}[\mathsf{Isometry}]$ representative $V : X \to X' \otimes G$ exposes $\rho \mapsto \mathrm{tr}_G(V\rho V^\dagger)$. Isometric relabelings of G preserve this map and generate the equivalence on representatives [30, Eqs. (3)–(4), Thm. 22].

To allocate a counter at a chosen value, we first need a type with K basis values. Write $[0] = 0$, $[1] = 1$ and $[k+1] = 1 + [k]$ for $k \geq 1$. The singleton value is $*$. In a larger counter, zero is left $*$ and $j+1$ is right j. The body of count_K supplies the return type through its right summand. For $0 \leq j < K$, choose a permutation $p_j : a + ([K-1] \otimes a) \leftrightarrow [K] \otimes a$ that sends left x to $\langle j, x\rangle$. Lifting p_j then gives alloc_j. For the basis copier on inhabited b, the heap $[|\mathscr{B}(b)| - 1] \otimes b$ supplies the $|\mathscr{B}(b)|^2 - |\mathscr{B}(b)|$ inputs needed to complete the diagonal injection to a permutation.

An embedded unitary acts by the same map. Embedding an isometry V gives $\mathscr{E}_V(\rho) = V\rho V^\dagger$, where the matrix subscript distinguishes conjugation from syntax elaboration in the environment Σ. To form count_K, apply $I_{[K]} \otimes \beta$ and distribute the counter over the two tags. Apply S_K on the returning tag alone. The blocks are now $I \otimes A, I \otimes B, S_K \otimes C, S_K \otimes D$, so exit k is $S_K^k \otimes x_k$. Closing $[K] \otimes c$ and preparing the counter at zero gives (6.1).

The formulas group outputs by counter value, so hiding the counter uses the swap in Section 6 to move it to the right. Executable `count K beta` returns $b \otimes [K]$. To keep the counter first, follow

(a) $\mathscr{R}[\mathscr{C}]$: a heap enters beside the input, and only access to it counts



(b) $\mathscr{L}[\mathscr{C}]$: garbage leaves beside the output, and its contents do not count

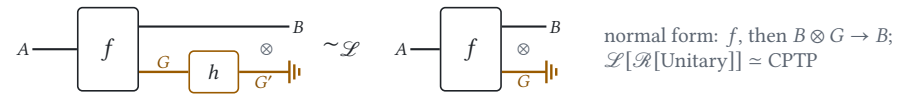


Fig. 14. Allocation exposes an input summand of a larger reversible map. Hiding exposes one output factor of an isometry. The unused input is zero, so the unitary $h : H' \to H$ preserves it. The isometry $h : G \to G'$ relabels the discarded output. Both preserve the exposed map. Here $\mathscr{C}$ = Unitary in $\mathscr{R}[\mathscr{C}]$ and $\mathscr{C}$ = Isometry in $\mathscr{L}[\mathscr{C}]$.

it with `swap`. To hide it, follow it with `discard`. On $b \otimes g$ the latter channel is $(I_b \otimes \mathsf{discard}_g); \mathsf{unit}_\otimes$, where $\mathsf{discard}_g : g \nrightarrow 1$ traces out the whole register.

For the empty type, $\mathsf{discard}_0$ is obtained by hiding the unit factor of the unique isometry $0 \rightarrowtail 1 \otimes 1$, constructed by lift from $0 + 1 \leftrightarrow 1 \otimes 1$. Its action is the unique linear map from $\mathrm{Op}(0)$ to $\mathrm{Op}(1)$. Set $\mathsf{measure}_0$ to the embedded identity on 0. These supply the empty cases omitted from the inhabited-type formulas in the body.

For while, allocate a fresh bit at 0 and apply the ordinary iso

$$\omega_{\mathrm{tag}} = \{ \begin{array}{l} \langle \mathrm{exit}\, x, 0\rangle \leftrightarrow \langle \mathrm{exit}\, x, 0\rangle, \\ \langle \mathrm{exit}\, x, 1\rangle \leftrightarrow \langle \mathrm{exit}\, x, 1\rangle, \\ \langle \mathrm{back}\, y, 0\rangle \leftrightarrow \langle \mathrm{back}\, y, 1\rangle, \\ \langle \mathrm{back}\, y, 1\rangle \leftrightarrow \langle \mathrm{back}\, y, 0\rangle \quad \} \end{array} : (b + c) \otimes \texttt{Qubit} \leftrightarrow (b + c) \otimes \texttt{Qubit}.$$

The clauses permute the basis. Fixing the fresh bit at 0 gives the tag isometry T. On $v \in [\![b]\!]$ and $y \in [\![c]\!]$ it acts by

$$T(v, y) = (\iota_{\mathrm{ext}} v) \otimes |0\rangle + (\iota_{\mathrm{ret}} y) \otimes |1\rangle.$$

Applying T after U and tracing out the copied bit removes the cross terms between the exit and return tags. We can therefore keep the exit block and feed back the return block by the classical sum. This is the operation we call while. Because T copies only the tag, the two payload types may differ.

When $a = b = c$, let $\mathsf{factor} : a + a \leftrightarrow a \otimes \texttt{Qubit}$ and $\mathsf{dist} = \mathsf{factor}^\dagger$, and put $\hat{\beta} = \mathsf{dist}; \beta; \mathsf{factor}$. Thus $F_g = \mathsf{factor}; \sigma_{a,\texttt{Qubit}}$ places the guard first, as does `uniti + uniti; factor` in the executable language. Writing $E(\omega) = \mathsf{arr}(\mathsf{arr}\,\omega)$, with $[\![E(\omega)]\!] = \mathscr{E}_{[\![\omega]\!]}$, the open call in Figure 9(a), written with the payload first, is

$$E(\mathsf{factor}); E(\hat{\beta}); (I_a \otimes \mathsf{measure}_{\texttt{Qubit}}); E(\mathsf{dist}).$$

The first factor; dist cancel, leaving the original body followed by exposure, measurement and restoration of its tag.

Soundness. Induct jointly on declaration-prefix length and source-subprogram size as in the static rules. For each basis input the matching clause supplies a column. At a local call the induction hypothesis gives a unitary for its checked iso, so any enclosed feedback has a unique external

output by Theorem 4.1. The table denotes a unitary because the admission rule requires these columns to form an orthonormal basis of the output. Sequence, sum, tensor and dagger preserve unitarity, as does trace by Theorem 4.1.

For lift, sending u to $(u, 0)$ preserves inner products, as does the unitary that follows. The same argument applies to count_K, which sends u to $|0\rangle \otimes u$ before applying the enlarged loop. Embedding an isometry gives a channel, and tracing out one output preserves that property. For `while`, each partial sum in (6.2) is completely positive because each exit acts by $\rho \mapsto x_k \rho x_k^\dagger$. The cone is closed in finite dimension, so the limit is completely positive too. The limit preserves trace because the exit probabilities sum to one on every unit input, by $\sum_k x_k^\dagger x_k = I$.

These cases establish that evaluation computes the denoted unitary, isometry or channel at its respective layer. A closed term $t : a$ denotes a vector in $\mathbb{C}^{\mathscr{B}(a)}$ whose norm depends on its coefficients. Conversely, the columns of any finite unitary V give the admitted clauses $s \leftrightarrow \sum_t V_{ts} t$, including the empty table.

B Convergence, probability and elimination

For a classical reversible body, follow an external input through its permutation table. If the path entered an internal cycle, the first value it reached there would have two predecessors. It can therefore visit each return value at most once and must exit within $|\mathscr{B}(c)| + 1$ calls, as in reversible flowchart semantics [23, Prop. 3.10] [36, Prop. 5.3]. In Figure 15 the two external paths give a swap. The split illustrated by `extraPort` in Section 4 lets us bound the weight left after each call.

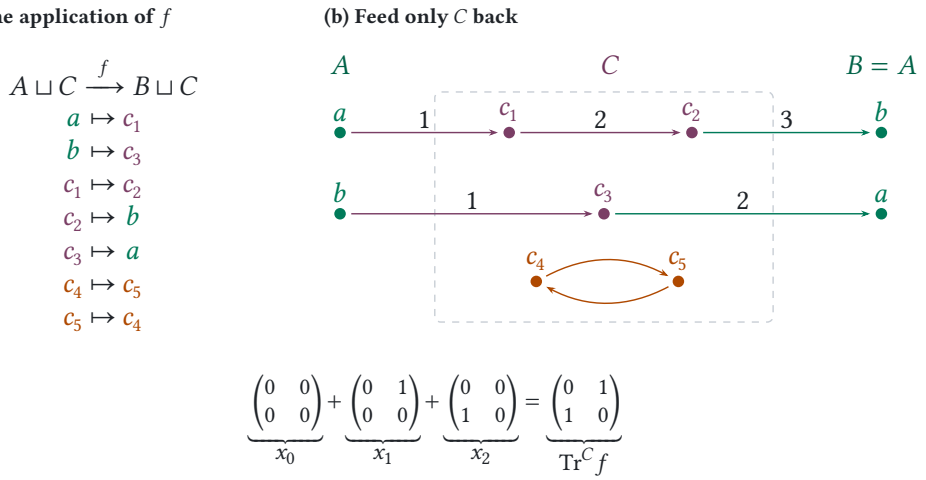


Fig. 15. A reversible loop on basis sets $A = \mathscr{B}(a)$, $B = \mathscr{B}(b)$ and $C = \mathscr{B}(c)$ has two external paths and an internal cycle. An edge numbered $k + 1$ is the call after k returns. The two nonzero exit maps have disjoint domains and their sum is a swap.

Proof of Theorem 4.1. If we start with y on the return wire, the next call emits By and sends Dy back. After j more returns the amplitude that can leave is $BD^j y$, so the states that never emit form $Y_0 = \bigcap_{j \geq 0} \ker BD^j$, the unitary part of D in the Sz.-Nagy–Foiaş decomposition [3, Thm. 3.3.7].

For $y \in Y_0$ the whole output of U is $(0, Dy)$, with $Dy \in Y_0$. Preservation of norm makes D an isometry on Y_0, hence onto there by finite dimensionality. Since U carries $0 \oplus Y_0$ onto itself, its restriction there is a unitary D_0 whose eigenvectors span Y_0. Conversely, if $Dv = \lambda v$ with $|\lambda| = 1$, norm preservation gives $Bv = 0$, hence $BD^j v = 0$ for every j. This identifies the two descriptions of

Y_0. On the orthogonal complement, U restricts to the unitary U_1 that takes $X \oplus Y_1$ onto $X' \oplus Y_1$, where $Y_1 = Y_0^{\perp}$. In particular, the first return Cu lies in Y_1 and D keeps every later return there.

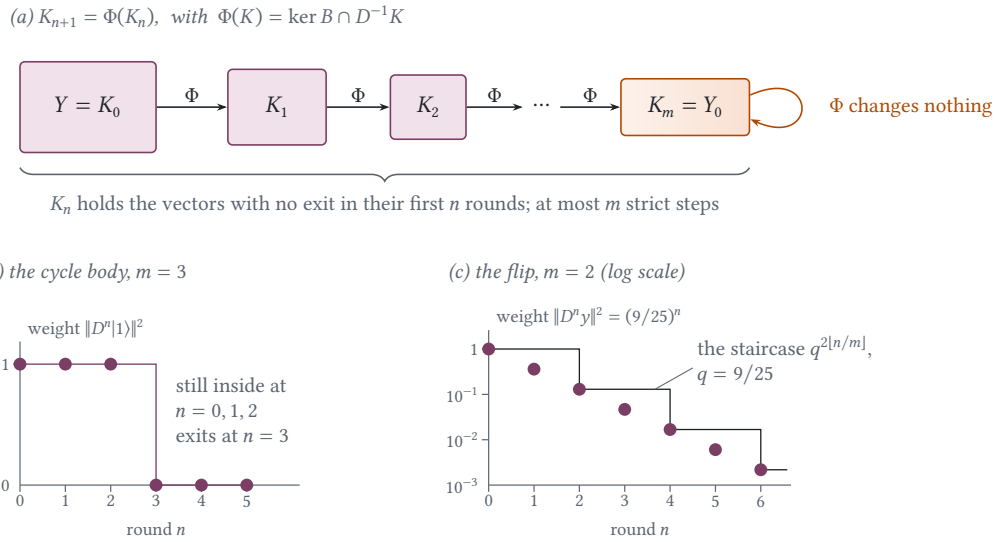


Fig. 16. The no-emission spaces stabilize within $\dim Y$ steps. Panel (b) is the four-state cycle $0 \mapsto 1 \mapsto 2 \mapsto 3 \mapsto 0$, with external state 0. Its three-dimensional return block has $D|1\rangle = |2\rangle$, $D|2\rangle = |3\rangle$ and $D|3\rangle = 0$. In panel (c), $\|D_1\| = 3/5$ and $m = 2$, so $q = \|D_1^2\| = 9/25$. Squaring the norm bound gives the weight bound. Each pair of calls multiplies the weight by q^2, so the staircase $q^{2\lfloor n/m \rfloor}$ meets the exact weight $(9/25)^n$ at every even n.

To see how quickly amplitude leaves Y_1, let K_j consist of the states that emit nothing for the first j calls. Then $K_0 = Y$ and $K_{j+1} = \{y : By = 0,\ Dy \in K_j\}$. Each space lies inside the one before it. If two successive spaces agree, the recurrence gives the same space at every later call. There can be at most $m = \dim Y$ strict decreases, so $K_m = Y_0$ (Figure 16).

Write $D_1 = D|_{Y_1}$ and $B_1 = B|_{Y_1}$. If $Y_1 = 0$, then $C = 0$, every delayed exit vanishes and we can take $F = 0$. Otherwise $\dim Y_1 > 0$ and hence $m > 0$. A unit vector $y \in Y_1$ must emit on at least one of its first m calls, since it lies outside K_m. Conservation of norm gives

$$1 - \|D_1^m y\|^2 = \sum_{j=0}^{m-1} \|BD^j y\|^2 > 0.$$

The unit sphere is compact, so even the largest remaining norm $q = \|D_1^m\|$ is less than one. After $k = pm + r$ calls, with $0 \le r < m$, the first p groups of m calls leave norm at most q^p, and the last r calls cannot increase it. This bounds $\|D_1^k\|$ by q^p.

By this bound the sum $F = \sum_{j \ge 0} D_1^j C$ converges. Its first term is C and the remaining terms sum to $D_1 F$, so $F = C + D_1 F$. Applying B_1 sends each return to its exit. These delayed exits are absolutely summable even with a polynomial weight in j. Adding the immediate exit gives $L = A + BF$. If two solutions differ by w, subtracting their wire equations gives $Dw = w$. On input $(0, w)$ the body therefore outputs (Bw, w). Since the returning component has the full input norm, $Bw = 0$ and the two solutions give the same external output.

For the solution Fu, the body sends (u, Fu) to (Lu, Fu). Canceling $\|Fu\|^2$ from the two squared norms gives $\|Lu\| = \|u\|$. The external spaces have the same finite dimension because U is unitary, so L is onto as well.

To account for the probability, let $T_j = C^\dagger D^{\dagger j} D^j C$. On a unit input u, $\langle u, T_j u\rangle$ is the probability still inside after call $j+1$. The first call gives $x_0^\dagger x_0 = I - T_0$. On later calls the weight that leaves is exactly the drop in the weight still inside, since

$$x_{j+1}^\dagger x_{j+1} = C^\dagger D^{\dagger j} \underbrace{B^\dagger B}_{I - D^\dagger D} D^j C = T_j - T_{j+1}.$$

Adding through exit k gives $\sum_{j=0}^{k} x_j^\dagger x_j = I_X - T_k$. The residual term tends to zero by the decay just proved, giving total exit probability one even for a body with infinitely many possible exits. □

A state can stay inside forever without being fixed. For example, if $Dv = -v$ then every call preserves its norm, so $Bv = 0$. Only a nonzero state with $Dv = v$ makes $I - D$ singular at phase one.

B.1 Eliminating a return value

For a self-entry different from one, the single-port reduction is the scattering update [25, § 4, Thm. 12] [52, App. A, Eq. (A1)].

Proof of Proposition 4.2. Choose the returning basis value q we want to remove, and leave the other return wires open for this step. Write $Q = \mathbb{C}q$ and $Y_{\text{rest}} = Y \cap Q^\perp$. The ports left open form $E_{\text{in}} = X \oplus Y_{\text{rest}}$ and $E_{\text{out}} = X' \oplus Y_{\text{rest}}$. In this order the body has matrix

$$\begin{pmatrix} M & b_q \\ c_q & d_q \end{pmatrix}, \qquad M : E_{\text{in}} \to E_{\text{out}}, \quad b_q : Q \to E_{\text{out}}, \quad c_q : E_{\text{in}} \to Q, \quad d_q = D_{qq}.$$

On input $w = (u, y_{\text{rest}})$, the other ports send amplitude $c_q w = (Cu)_q + \sum_{r \neq q} D_{qr} y_r$ to q. If λ comes back on q itself, the same port receives a further $d_q \lambda$. Closing this one wire therefore asks us to solve $\lambda = c_q w + d_q \lambda$. The other ports receive $Mw + b_q \lambda$.

If $d_q \neq 1$, substitution gives the new table $M + b_q c_q/(1 - d_q)$, the Schur complement of $d_q - 1$ in the matrix with blocks $M, b_q, c_q, d_q - 1$. This division also works for a phase such as -1 or i. When $d_q = 1$, that entry accounts for the whole squared norm of its row and column, so $c_q = b_q = 0$. The wire equation reduces to $\lambda = \lambda$, and the other ports see Mw for every choice of λ. In either case the new table is $\mathrm{Tr}^Q(U)$, which is unitary by Theorem 4.1.

For `guardRound` there is only one returning value. Substituting $M = d_q = 3/5$, $b_q = -4/5$ and $c_q = 4/5$ gives the single coefficient $3/5 - (4/5)^2/(1 - 3/5) = -1$ of the clause in Section 4.3. In `flipRound`, the updated table still contains `back 1`; its new self-entry $9/25$ is the d_q for the next step (Figure 4).

A solution on all the wires restricts to a solution on the remaining wires by the substitution just made. Conversely, from a solution on those wires we recover λ by the same division when $d_q \neq 1$ and may choose $\lambda = 0$ when $d_q = 1$, since $b_q = 0$ and $c_q = 0$. Thus each step preserves the external output. Each step also removes one basis value and uses only field operations with a nonzero divisor. In any order there are $|\mathcal{B}(c)|$ steps, ending with a table over the original field for the gate L. The same argument applies after putting a phase on the return wire, since the modified body is again unitary. □

Closed sums for the running rotation. In `guardRound`, $A = D = t = 3/5$, $C = s = 4/5$ and $B = -s$, so $x_0 = t$ and $x_{k+1} = -s^2 t^k$. After n calls on unit input, let r_n denote the remaining return amplitude,

ℓ_n the sum emitted, and p_n the measured weight emitted. Summing the two geometric series gives

$$r_n = st^{n-1},$$
$$\ell_n = t - s^2 \sum_{j=0}^{n-2} t^j = -1 + (1+t)t^{n-1},$$
$$p_n = t^2 + s^4 \sum_{j=0}^{n-2} t^{2j} = 1 - s^2 t^{2(n-1)}.$$

Call n includes $n-1$ trips (Section 5), so the sums are empty for $n = 1$. In Table 2, interference makes $|\ell_n|^2$ fall and rise, while p_n only adds nonnegative probabilities.

	one call		coherent feedback		guard measured
call n	emitted x_{n-1}	still returning r_n	sum emitted ℓ_n	squared length $\lvert\ell_n\rvert^2$	halted so far p_n
1	$\frac{3}{5}$	$\frac{4}{5}$	$\frac{3}{5}$	0.3600	0.3600
2	$-\frac{16}{25}$	$\frac{12}{25}$	$-\frac{1}{25}$	0.0016	0.7696
3	$-\frac{48}{125}$	$\frac{36}{125}$	$-\frac{53}{125}$	0.1798	0.9171
4	$-\frac{144}{625}$	$\frac{108}{625}$	$-\frac{409}{625}$	0.4282	0.9701
5	$-\frac{432}{3125}$	$\frac{324}{3125}$	$-\frac{2477}{3125}$	0.6283	0.9893
$n \to \infty$	0	0	-1	1	1

Table 2. `guardRound` on $|*\rangle$. Coherent feedback adds exits, while guard measurement adds their squared magnitudes. A record keeps exit x_{n-1} with label $n-1$. Only p_n is a halting probability; decimal entries are rounded to four places.

A table and an unknown gate. For a scalar rotation with cosine $-1 \le t < 1$, the trace is $t - (1-t^2)/(1-t) = -1$. At the identity rotation it is $+1$. With a bypass and a final Hadamard, the probability of reading 1 therefore jumps from one to zero at the identity. In a fixed circuit using finitely many calls to the supplied gate, each amplitude is a finite sum of products of gate entries, so the probability varies continuously with the rotation angle. If it gives probability p at the identity, its error there is p and its error on nearby rotations tends to $1-p$. The worst-case error is therefore at least $\max(p, 1-p) \ge 1/2$. Given the clause table, elimination can test $t = 1$ exactly [20, Remark B.1]. Section 5 follows the light from its first trip around an empty ring.

C Records, measurement and phase

On a return, the bit flip in `count 2 guardRound` becomes the cyclic successor S_K for a K-state counter. Distributivity $d_K : [K] \otimes (a + c) \leftrightarrow ([K] \otimes a) + ([K] \otimes c)$ and its output counterpart d'_K give the body of Figure 7.

$$\beta_K = d_K^{-1} ; (I_{[K]} \otimes \beta) ; d'_K ; (I_{[K]\otimes b} \oplus (S_K \otimes I_c)).$$

The upper row of its matrix sends amplitude out with the current label. The lower row applies S_K before sending amplitude back.

$$W_K = [\![\beta_K]\!] = \begin{pmatrix} I_K \otimes A & I_K \otimes B \\ S_K \otimes C & S_K \otimes D \end{pmatrix}. \qquad R_K u = (\mathrm{Tr}^{\mathbb{C}^K \otimes Y} W_K)(|0\rangle \otimes u).$$

When both ports carry one type P with $[\![P]\!] = X = X' = Y$, the sheet diagram draws the guard and data on one sheet. The change of coordinates $F_g = \mathrm{factor}; \sigma_{P,\mathrm{Qubit}} : P + P \leftrightarrow \mathtt{Qubit} \otimes P$ puts

the guard first, giving $\widehat{U}_g = [\![F_g]\!]U[\![F_g]\!]^{-1}$. With a counter, $F_{K,g} = d_K^{-1};(I_{[K]} \otimes F_g)$ puts counter, guard and data in that order. Its inverse restores the two ports.

On the first call, $(|0\rangle \otimes u, 0)$ has a prepared counter and no returning amplitude. The outputs are $|0\rangle \otimes Au$ and $|1 \bmod K\rangle \otimes Cu$. Figure 17 shows how the same outgoing guard controls a count or a fresh record.



Fig. 17. One call to the same body U, followed by an operation controlled on the outgoing guard $g = 1$. A filled dot denotes a coherent control. The left circuit carries j into the next round. In the right circuit the discarded bit distinguishes the tags without distinguishing states of the payload q.

From a finite counter to the exact count. Give each return count its own basis state in $H_{\mathrm{rec}} = \ell^2(\mathbb{N})$. With the counter first, $Ru = \sum_k |k\rangle \otimes x_k u$ has squared norm $\sum_k \|x_k u\|^2 = \|u\|^2$. For `guardRound` on $|*\rangle$ the data factor is always $|*\rangle$, and the count state is $\xi_{\mathrm{rot}} = t|0\rangle - s^2 \sum_{k\geq 1} t^{k-1}|k\rangle$, with $t = 3/5$ and $s = 4/5$. Section 7 factors out $L|*\rangle = -|*\rangle$, so its count vector is $-\xi_{\mathrm{rot}}$. In the bypass experiment we factor out $|1\rangle$ on the rotating path, so its count state is ξ_{rot}.

In `bypassRound`, the bypass exits at once, so $R|0\rangle = |0\rangle \otimes |0\rangle$, while $R|1\rangle = \xi_{\mathrm{rot}} \otimes |1\rangle$. Only the rotating arm has a nonzero $|1\rangle$ count amplitude, so $R|+\rangle$ is entangled. With a K-state counter, the rotating arm's amplitude at label zero is $g_0 = t - s^2 t^{K-1}/(1 - t^K)$, since all positive multiples of K reach that label. The bypass keeps $|0\rangle$, so g_0 is the overlap of the two unit count states. At output 1 the final Hadamard gives half their difference. Its squared norm is $(2 - 2g_0)/4 = (1 - g_0)/2$. For $t = 3/5$ this gives one at $K = 1$ and $1/2$ at $K = 2$. Since $g_0 < t$ and $g_0 \to t$, it approaches the measured value $(1 - t)/2 = 1/5$ from above.

Projecting the counter onto $K^{-1/2} \sum_{j<K} |j\rangle$ gives $K^{-1/2} \sum_k x_k u = Lu/\sqrt{K}$, with probability $1/K$ on a unit input. For a known body, reversing the enlarged loop instead restores $|0\rangle \otimes u$ before applying L (Section 6).

Embed the K counter labels as the first K basis states by $J_K : \mathbb{C}^K \to H_{\mathrm{rec}}$. The finite and exact records agree term by term until a label wraps. Their difference is bounded by the remaining exits.

$$\|(J_K \otimes I)R_K - R\| \;\leq\; \sum_{k\geq K} \big\||k \bmod K\rangle - |k\rangle\big\| \, \|x_k\| \;\leq\; \sqrt{2} \sum_{k\geq K} \|x_k\|.$$

For $k \geq K$ the two labels are orthogonal unit vectors, at distance $\sqrt{2}$. Theorem 4.1 gives absolute convergence, so this tail tends to zero.

Square summability alone gives no such control over folding. For example, a record with amplitude $1/\sqrt{n}$ at each of $0, K, \ldots, (n-1)K$ has norm one, but folding modulo K adds all n amplitudes at zero and gives $\sqrt{n}|0\rangle$. Summing at phase one likewise gives $\sqrt{n}$. As n grows, this shows why neither operation is bounded on the whole record space ℓ^2.

The exact count can also be written as a sequence of guard outcomes. In Ying's free coin histories, k returns followed by an exit give the word $J|k\rangle = |1^k 0\rangle$ in the $(k+1)$-coin sector [58, Eq. (34)]. These words are orthonormal, so J identifies the exact register isometrically with their closed span. For matched exit maps, $(J \otimes I)Ru$ and Ru give the same data when we discard the record. This comparison concerns completed histories. Ying's principal-system semantics also specifies an

initial coin state and uses the symmetric semantics [58, Def. 9]. Assolini and Di Pierro keep a qubit from each test. Their unitary approximations can fail to converge while a branch keeps running, although the output on completed branches converges for each input [4, § 4.1, Thm. 5.9].

A clock can also keep the count. One tick of delay per return puts $x_0u, x_1u, \dots$ in consecutive, orthogonal output slots [3, § 3.4]. At a common observation time, a shared sequence of post-halt ancilla states can likewise distinguish halt times. Branch-dependent sequences can let those histories interfere again [38]. For an exact record, the labels must still be orthogonal when observed. In the ring, a short pulse separates the exits in time, while steady light at one frequency brings them to the same mode (Appendix H).

What remains after hiding the count. For an input of the form $|u\rangle\langle v|$, tracing the exact record over its orthonormal basis leaves $\sum_k x_k|u\rangle\langle v|x_k^\dagger$. Such operators span the input matrix algebra, giving (6.2). For `bypassRound`, all delayed exits are multiples of $P = |1\rangle\langle 1|$. Their squared coefficients sum to $s^4/(1-t^2) = s^2$. Thus $u \mapsto |0\rangle \otimes Au + |1\rangle \otimes sPu$ gives the same measured channel when we hide the bit. The bit labels two Kraus operators, A on an immediate exit and sP on all delayed exits together.

For a finite counter put $G_j = \sum_{k \equiv j \pmod K} x_k$. Discarding that counter instead gives $\sum_{j<K} G_j \rho G_j^\dagger$. As a result, unequal counts congruent modulo K still interfere. At the Kth roots of unity, z^k depends only on k mod K, so averaging against $\bar{z}^{\,j}$ selects the entire residue class j. Averaging over the whole circle selects a single count. The exit series converges absolutely, so we may average it term by term to obtain

$$G_j = \frac{1}{K} \sum_{z^K=1} \bar{z}^{\,j} \Theta(z), \qquad x_k = \frac{1}{2\pi} \int_0^{2\pi} e^{-ik\varphi} \Theta(e^{i\varphi})\, d\varphi.$$

At phase one the exits add to L. Averaging $\Theta(z)\rho\Theta(z)^\dagger$ over the circle cancels every term with unequal counts, giving (8.1) by absolute convergence. Over the Kth roots, a pair of exits k, j instead acquires the factor $K^{-1}\sum_{z^K=1} z^{k-j}$. It is one if $k \equiv j \pmod K$ and zero otherwise, so the average is $\sum_{r<K} G_r \rho G_r^\dagger$, the data channel of the hidden K-counter. If $D^n = 0$, only counts $0, \dots, n$ can occur, so $n+1$ equally spaced phases give the measured loop exactly.

We can recover the measured channel by closing the return block of a single measured call, as in Figure 10. Let $\theta(\rho, \sigma) = \operatorname{diag}(\rho, \sigma)$ insert the two operator blocks and let φ extract the diagonal output blocks. Then $\Phi_U = \varphi \circ \mathscr{C}_U \circ \theta$ is the branching map of Section 6 [3, Def. 3.3.18]. In CPTR, the category of finite-dimensional C^*-algebras and completely positive trace-nonincreasing maps, the classical-sum trace $\mathrm{Tr}_{\mathrm{cl}}^{\mathrm{End}(Y)}(\Phi_U)$ sends ρ to $\sum_k x_k \rho x_k^\dagger$ [49, § 5.1, Eq. (1)] [3, Thm. 3.2.11, Prop. 3.2.17].

It is the extraction of the diagonal blocks after each call that removes coherence between exit and return. Embedding the unitary in finite isometries and tracing there still gives the coherent loop [3, Cor. 3.3.16].

Relation to monitored first returns. The published test-first loop measures before applying U [59, § 3]. To obtain our body-first convention when $X' = X$, apply U once before entering it. Let $\iota_X u = (u, 0)$ and $P = \iota_X \iota_X^\dagger$. For $n \geq 1$, the exit map on call n is $\iota_X^\dagger U((I-P)U)^{n-1}\iota_X = x_{n-1} : X \to X$. Thus the monitored-return count includes the initial call and is one more than our number of returns [11, § 2].

To see where the least fixed point enters, consider the state σ just before the test. The exit contributes $P\sigma P$. On a return, the next call prepares $V\sigma V^\dagger$ for another test, where $V = U(I-P)$. Thus Ying's unfolding sends a map G to $\sigma \mapsto P\sigma P + G(V\sigma V^\dagger)$ [57, arXiv v1, Prop. 5.1(6), Cor. 5.1].

Starting from zero, n unfoldings give the first n exits, $\sum_{j=0}^{n-1} PV^j\sigma(V^j)^\dagger P$. For positive σ each new exit adds a positive operator. Since $V^\dagger V = I - P$, their total trace is $\operatorname{tr}\sigma - \operatorname{tr}(V^n\sigma(V^n)^\dagger)$. The partial sums are thus increasing with bounded trace, so they converge in finite dimension. Passing to the limit in the recurrence gives a solution. It is the least one because any positive solution bounds the initial zero and, by order preservation at each unfolding, every partial sum. Starting with $U\iota_X\rho\iota_X^\dagger U^\dagger$ and reading the external block gives $\sum_k x_k\rho x_k^\dagger = \mathscr{M}(\rho)$. In quantum Hoare logic the output trace accounts for termination [57]. Here $\sum_k x_k^\dagger x_k = I$ gives termination with probability one, while Theorem 7.1 decides the further question of which data the loop returns.

D Agreement and replacement

Proof of Theorem 7.1. Suppose the measured and coherent loops give the same channel. On a unit input u the coherent loop returns Lu, so a test for any unit vector w orthogonal to Lu has probability zero. The same test on the measured loop gives $\sum_k |\langle w, x_k u\rangle|^2$. For this sum to vanish, each term must vanish. Thus every $x_k u$ lies along Lu.

So far the multiplier could depend on the input. For a fixed exit k, put $V = L^\dagger x_k$, so that Vu lies along u for every u. In particular, $Ve_i = \lambda_i e_i$ on each basis vector. On $e_i + e_j$ this gives

$$V(e_i + e_j) = \lambda_i e_i + \lambda_j e_j = \lambda_{ij}(e_i + e_j).$$

Comparing the two coordinates gives $\lambda_i = \lambda_j$. Thus $V = b_k I$ and $x_k = b_k L$ for one scalar b_k on the whole input space. In dimension one there is only one coefficient to begin with.

What stays in the count register now follows from the two ways of summing exits. Since $\sum_k x_k = L \neq 0$, absolute convergence gives $\sum_k b_k = 1$. Since $\sum_k x_k^\dagger x_k = I$, we also have $\sum_k |b_k|^2 = 1$. The exact counter therefore returns a unit vector $\xi = \sum_k b_k|k\rangle$, the same for every input, and $Ru = \xi \otimes Lu$. Conversely, discarding such a factor gives $\mathscr{E}_L$. Taking the kth coordinate recovers $x_k = \langle k|\xi\rangle L$ for every exit.

It remains to decide this condition from finitely many exits. The immediate exit is $x_0 = A$. For $m = \dim Y > 0$, the next m exits are $BC, BDC, \ldots, BD^{m-1}C$. By Cayley–Hamilton, D^m is a linear combination of $I, D, \ldots, D^{m-1}$, and multiplying by D gives the same recurrence for all higher powers. Hence every later exit lies in the span of those m delayed exits. When $m = 0$, only the immediate exit can be nonzero.

The span of $x_0, \ldots, x_m$ thus contains every exit and, being finite-dimensional, also their sum L. On a nonzero input space L is nonzero. If this span has dimension at most one, it must be the line through L, so every x_k is a multiple of L. Conversely, multiples of L pass the finite test. □

The dense cost in Section 7 follows by forming BW and updating W to DW for each of the m delayed exits.

We can also compute the measured channel directly. Restrict B, D to the emitting space Y_1 as in Appendix B, and write $\mathscr{E}_T(\rho) = T\rho T^\dagger$. On return density matrices the next call acts by $\mathscr{E}_{D_1}$. Its powers decay, so summing the delayed exits gives

$$\mathscr{M} = \mathscr{E}_A + \mathscr{E}_{B_1}(I - \mathscr{E}_{D_1})^{-1}\mathscr{E}_C.$$

Comparing this with $\mathscr{E}_L$ on matrix units is a second finite test. The rational comparison solves the full system $(I - \mathscr{E}_D)S = \mathscr{E}_C$ on all matrix units by row reduction, sets free coordinates to zero and checks the residual. A solution exists by the decaying series on Y_1. Any difference T between solutions satisfies $T(\rho) = D^jT(\rho)(D^\dagger)^j$ for every j. Only its Y_0-to-Y_0 block survives as j grows, and B kills that block. Thus $\mathscr{E}_A + \mathscr{E}_B S$ is independent of the free coordinates. The solve uses $O(m^6 + m^4n^2)$ field operations before forming its n^2-by-n^2 output.

The span test needs only the first nonzero exit P and its scalar multiples. Its normalization also stays in the coefficient field $\mathbb{F}$: the entries of $U^\dagger = U^{-1}$ lie in $\mathbb{F}$, so the conjugates of the entries of P do too. Thus the squared column norm $q = (P^\dagger P)_{11}$ belongs to $\mathbb{F}$.

For two loops with separate exact counts, forgetting both records composes their measured channels. With one shared exact count, exits with the same total add before we forget it, so we apply the formula to the combined body of Section 8. Comparing these channels on matrix units decides whether sharing changes the data.

The bypass shows why we test proportionality exactly. As t approaches one from below, $x_1 = -(1-t^2)|1\rangle\langle 1|$ tends to zero. The delayed terms sum to $-(1-t^2)/(1-t) = -(1+t)$ on that branch, so the coherent caller still reads 1 with certainty, while measuring the guard gives 1 with probability $(1-t)/2$. As that probability tends to zero, rounding x_1 to a zero multiple of x_0 would accept a change the caller can almost certainly detect.

Channel contexts. Fix the hole type $a \rightsquigarrow b$ and abbreviate it by $\mathscr{H}$ to the left of a context judgment. For fixed closed channels h, h_j, the contexts are generated by

$$\begin{aligned}\mathscr{C} \;::=\; & [-] \mid h;\mathscr{C} \mid \mathscr{C};h \mid h \otimes \mathscr{C} \mid \mathscr{C} \otimes h \\ & \mid \mathsf{case}\big(\Psi_i;\mathscr{C},(\Psi_j;h_j)_{j\neq i}\big).\end{aligned}$$

The rules for the hole, sequence and tensor are

$$\mathscr{H} \vdash [-] : a \rightsquigarrow b \qquad \frac{h : e \rightsquigarrow c \qquad \mathscr{H} \vdash \mathscr{C} : c \rightsquigarrow d}{\mathscr{H} \vdash h;\mathscr{C} : e \rightsquigarrow d} \qquad \frac{\mathscr{H} \vdash \mathscr{C} : c \rightsquigarrow d \qquad h : d \rightsquigarrow e}{\mathscr{H} \vdash \mathscr{C};h : c \rightsquigarrow e}$$

$$\frac{h : e \rightsquigarrow f \qquad \mathscr{H} \vdash \mathscr{C} : c \rightsquigarrow d}{\mathscr{H} \vdash h \otimes \mathscr{C} : e \otimes c \rightsquigarrow f \otimes d} \qquad \frac{\mathscr{H} \vdash \mathscr{C} : c \rightsquigarrow d \qquad h : e \rightsquigarrow f}{\mathscr{H} \vdash \mathscr{C} \otimes h : c \otimes e \rightsquigarrow d \otimes f}.$$

For a finite instrument $(\Psi_j : c \to c_j)_{j\in J}$, meaning $\Psi_j : \mathrm{End}([\![c]\!]) \to \mathrm{End}([\![c_j]\!])$, the maps are completely positive and their output traces sum to the input trace. Placing each output in its summand of $\bigoplus_j [\![c_j]\!]$ gives an instrument with a common output space. Its case rule is

$$\frac{i \in J \quad \mathscr{H} \vdash \mathscr{C} : c_i \rightsquigarrow d \quad h_j : c_j \rightsquigarrow d \;(j \neq i)}{\mathscr{H} \vdash \mathsf{case}\big(\Psi_i;\mathscr{C},(\Psi_j;h_j)_{j\neq i}\big) : c \rightsquigarrow d}.$$

Plugging h into the hole interprets sequencing by channel composition and tensor by the tensor product of channels. The case denotes $[\![\mathscr{C}[h]]\!] \circ \Psi_i + \sum_{j\neq i} [\![h_j]\!] \circ \Psi_j$. Thus exactly one syntactic occurrence depends on the hole, and each branch continuation has the channel type stated by the rules. To retain the outcome, let branch j embed its output in the jth summand of a common tagged output. A caller with finitely many occurrences can replace them one at a time because fixing the others leaves one of the single-hole contexts above. The hole accepts a channel, so coherent control of a chosen implementing iso is outside this grammar.

Proof of Corollary 7.2. Discarding the exact count removes every term with two different labels and leaves $\sum_k x_k \rho x_k^\dagger = \mathscr{M}(\rho)$. With a finite counter, exits whose counts agree modulo K carry the same label. Under the agreement condition, each contributes $b_k L u$. Factoring out the common data vector gives $R_K u = \xi_K \otimes Lu$, where $\xi_K = \sum_k b_k |k \bmod K\rangle$. For any unit input u, both $R_K u$ and Lu have norm one, so ξ_K does too. Discarding it gives $\mathscr{E}_L$. If $X = 0$, then $X' = 0$ and there is only one channel to compare.

Equal channels still agree when the caller keeps a finite reference: elementary tensor operators span the joint matrix algebra. They also agree after composition with a fixed channel or discard. In a case expression we can substitute equal channels on the unnormalized state of each outcome,

then add the outputs if the caller forgets the outcome. Induction on the context now gives the same observations for both loops.

For the converse we exhibit tests available to the caller. On either space we can prepare basis states and, for each pair r, s, the states $(3|r\rangle + 4|s\rangle)/5$ and $(3|r\rangle + 4i|s\rangle)/5$. Their projectors, together with the basis projectors, span the matrix algebra: the two superpositions supply the real and imaginary off-diagonal parts. Thus if the channels differ, one of these inputs gives different outputs. A basis measurement detects a difference on the diagonal. If the diagonals agree, projecting onto one of the two superpositions detects a nonzero real or imaginary part. Each preparation and test needs only a basis state or measurement and a two-level unitary over $\mathbb{Q}(i)$. The finite-counter converse follows from the residue-span calculation under How large a counter suffices below. □

Section 7 claims that replacement beside an identity bypass is safe exactly when every external input exits at once. To see why, let the bypass carry a nonzero space Z. The first exit is $I_Z \oplus A$, whereas each later exit $0 \oplus x_k$ vanishes on Z. The only scalar multiple of the first exit that does so is zero, so the exits span a line only if every delayed exit vanishes. Exit completeness then gives $A^\dagger A = I$, so $A^\dagger A + C^\dagger C = I$ forces $C = 0$. Conversely, when $C = 0$ no external input reaches the return wire, even if the body has an isolated return port.

A caller who acts only on the data gets the same result whether it discards the count before or after that action. Let H hold the count and τ be its joint state with the data. For each Kraus operator T of a branch on the data, $\mathrm{tr}_H((I_H \otimes T)\tau(I_H \otimes T)^\dagger) = T(\mathrm{tr}_H\,\tau)T^\dagger$. Summing over the branch's Kraus operators gives the same unnormalized output, and hence the same branch probability, in either order.

Checking compiled tables. For each real rational body below we compare both orders of elimination with a direct solution of the wire equations. All three give the same gate. Independently, we solve the measured channel equations on matrix units and compare that channel with the gate's channel. Its answer agrees with the finite exit test in every row of Table 3. Write $R = \left(\begin{smallmatrix} t & -s \\ s & t \end{smallmatrix}\right)$ with $t = 3/5$, $s = 4/5$. The reflection is $\left(\begin{smallmatrix} t & s \\ s & -t \end{smallmatrix}\right)$. The mixed isolated return is $(1 \oplus R)(R \oplus 1)(1 \oplus R)^{\mathsf{T}}$: it mixes the live and trapped return coordinates of `extraPort` after ordering the live one first. The last two families have body $R \otimes H$. For the dense reflection, $H = I_8 - \mathbf{1}\mathbf{1}^{\mathsf{T}}/4$ with $\mathbf{1} = (1, \ldots, 1)^{\mathsf{T}}$, so $x_0 = tH$ and $x_1 = -s^2 I$ are independent. For the independent rotations, $H = I_{16}$ and all exits are scalar multiples of I_{16}.

Table 3. Agreement on rational clause tables. Each dimension counts basis values.

body	dim X	dim Y	agree?	body	dim X	dim Y	agree?
`guardRound`	1	1	yes	`twoSections`	1	2	yes
`bypassRound`	2	1	no	`mixed isolated return`	1	2	yes
`flipRound`	2	2	no	`dense reflection`	8	8	no
`flipOnExit`	2	2	yes	`independent rotations`	16	16	yes
`reflection`	1	1	yes				

What a reset changes. Paetznick and Svore's Figure 9 starts the ancilla at zero. On success, outcome zero applies $V_3 = (I + 2iZ)/\sqrt{5}$ with probability 5/8. Failure leaves the data unchanged, so preparing zero repeats the same attempt [45, Fig. 9]. Feeding the failed guard straight back gives a different program. In the circuit matrix, A is proportional to V_3 and $C = \gamma I$, $|\gamma|^2 = 3/8$. The controlled signs and final data Z preserve each data basis value d, leaving a two-dimensional ancilla matrix U_d. Its five Hadamards contribute -1 to the determinant and the three T gates and one $T^\dagger$ contribute i. The two controlled signs have product determinant one, as does the final data phase $(-1)^d I_2$,

so $\det U_d = -i$ for both values. The unitary cofactor identity $(U_d)_{12} = -\det(U_d)\overline{(U_d)_{21}}$ now gives $B = i\bar{\gamma} I$. The immediate exit A and the next exit $BC = (3i/8)I$ already span two directions.

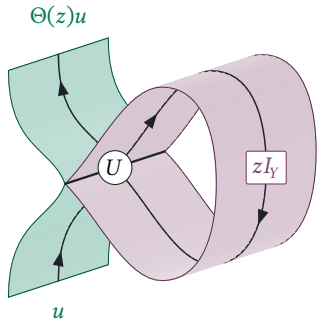
How large a counter suffices. For $K > d = \dim Y_1$, grouping exits by their count modulo K preserves their span. If $d = 0$ there is only the immediate exit. Otherwise $D_1^k \to 0$, so $I - D_1^K$ has a geometric-series inverse. Put $h_j = B_1 D_1^j (I - D_1^K)^{-1} C$. Expanding the inverse groups delayed exits K apart: $G_r = h_{r-1}$ for $1 \le r < K$, while $G_0 = A + h_{K-1}$ also contains the immediate exit.

Only G_0 mixes a delayed group with A. Since $K - 1 \ge d$, the first d groups $h_0, \dots, h_{d-1}$ occur alone as $G_1, \dots, G_d$. Cayley–Hamilton for D_1 puts every later h_j in their span. Subtracting successive groups recovers each delayed exit, $x_{j+1} = h_j - h_{j+K}$, and subtracting h_{K-1} from G_0 recovers A. Conversely each G_r is a convergent sum in the finite-dimensional span of the exits. Hiding the counter uses the G_r as Kraus operators. If their channel equals $\mathscr{E}_L$, the orthogonality argument of Theorem 7.1 gives $G_r \in \mathbb{C}L$ for each r, so every exit is on the same line. This proves the finite-counter converse.

Why a clock of size m can miss a difference. The output interface matters. If a caller reads the count, even a fixed count independent of the data is an observable output. Conversely, a finite clock may hide a failure of agreement. For $m > 0$, use the permutation that fixes `exit 0` and cycles through `exit` $1 \mapsto$ `back` $q_1 \mapsto \cdots \mapsto$ `back` $q_m \mapsto$ `exit` 1, with m distinct return values $q_1, \dots, q_m$. Its only nonzero exits are $x_0 = |0\rangle\langle 0|$ and $x_m = |1\rangle\langle 1|$. The m-clock assigns label 0 to both exits. Ignoring it preserves $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$, whereas a measurement on every return dephases $|+\rangle$ in that basis. Checking just this clock would accept the replacement incorrectly. Every returning basis value eventually exits, so here $d = m$: the bound $K > d$ is sharp.

E The derivative and its average

For `guardRound`, the first returns at the phase gate are s, stz, st^2z^2, so $F_z = s/(1 - tz)$ with $s = 4/5$ and $t = 3/5$. In general Theorem 4.1 puts $C(X)$ in the emitting space Y_1, where powers of D_1 decay geometrically. Thus $F_z = \sum_{k \ge 0} z^k D_1^k C = (I - zD_1)^{-1} C$ even when $I - zD$ is singular.



	$j = 0$	$j = 1$	$j = 2$
$k = 0$	T_{00}	$z^{-1}T_{01}$	$z^{-2}T_{02}$
$k = 1$	zT_{10}	T_{11}	$z^{-1}T_{12}$
$k = 2$	z^2T_{20}	zT_{21}	T_{22}

$$T_{kj} = x_k \rho x_j^\dagger$$

$$\Theta_{\mathrm{rot}}(z) = \frac{t - z}{1 - tz},$$

$$\Theta_{\mathrm{bypass}}(z) = \mathrm{diag}(1, \Theta_{\mathrm{rot}}(z)).$$

$$\Theta(z) = \sum_{k \ge 0} z^k x_k = A + zB_1(I - zD_1)^{-1}C, \qquad L = \Theta(1).$$

Fig. 18. Each return at left adds a factor z. At right, averaging $z^{k-j}T_{kj}$ over the circle keeps exactly the diagonal $k = j$. The two responses below the grid are those of `guardRound` and `bypassRound`.

Proof of Theorem 8.1. We use the same body at two return phases. On Y_1 the wire equation is $F_z = C + zD_1F_z$, so a call takes (u, zF_zu) to $(\Theta(z)u, F_zu)$. Comparing this call with one on (v, wF_wv) at phase w, we can equate the input and output inner products because U_1 is unitary. With the

inner product linear in its second argument, this gives

$$\langle v, u\rangle + \bar{w}z\langle F_w v, F_z u\rangle = \langle \Theta(w)v, \Theta(z)u\rangle + \langle F_w v, F_z u\rangle.$$

Collecting the two occurrences of the returning inner product gives $I - \Theta(w)^\dagger\Theta(z) = (1-\bar{w}z)F_w^\dagger F_z$. On the open unit disk this is the transfer identity for the characteristic function of the unitary colligation [12] [7, Eq. (4.10)].

The decay in Theorem 4.1 makes F_z and $\Theta(z)$ analytic on a disk extending beyond the unit circle. Fix $|w| = 1$ and differentiate the identity in z. At $z = w$ the factor $1 - \bar{w}z$ vanishes, taking the term with F_z' with it. This leaves $\Theta(w)^\dagger\Theta'(w) = \bar{w}F_w^\dagger F_w$. Since $dw/d\varphi = iw$, this gives $Q(w) = F_w^\dagger F_w$.

To average this weight over phase, expand $F_z = \sum_{j\geq 0} z^j D_1^j C$. In $F_z^\dagger F_z$ the term indexed by j, k has phase z^{k-j}. Integration around the circle keeps only terms with $j = k$, just as it does for the channel in Figure 18. The geometric bound permits termwise integration, so the average is $\sum_j T_j$, where $T_j = (D_1^j C)^\dagger D_1^j C$ is the weight still returning after call $j+1$.

An exit after k returns contributes to the returning weight on k calls. Weighting $x_{j+1}^\dagger x_{j+1} = T_j - T_{j+1}$ by $j+1$ and summing through $j = p-1$ gives $\sum_{k\leq p} k x_k^\dagger x_k = \sum_{j<p} T_j - pT_p$. Geometric decay sends pT_p to zero, so the phase average $\sum_j T_j$ equals N. On a unit input, its expectation gives the mean number of returns.

Taking traces gives $\operatorname{tr} N = \sum_j \operatorname{tr} T_j$. Writing $T_j = (D_1^j C)^\dagger (D_1^j C)$, we can move the first factor to the end under the trace. The product now acts on Y_1, where the return row gives $CC^\dagger = I - D_1 D_1^\dagger$. Substituting this difference cancels consecutive terms:

$$\sum_{j=0}^{p-1} D_1^j C C^\dagger D_1^{\dagger j} = I_{Y_1} - D_1^p D_1^{\dagger p} \longrightarrow I_{Y_1}.$$

Taking the trace gives $\operatorname{tr} N = \dim Y_1$. When $Y_1 = 0$, $F_z = 0$, so all three quantities vanish. □

For monitored first returns, identify the external output unitarily with the input X. An exit after k returns takes $k+1$ calls, so counting calls multiplies $\Theta(z)$ by one more z. The relative phase derivative of $z\Theta(z)$ is therefore $I + Q(z)$. For a single monitored state, Grünbaum et al. express its mean first return time as the winding of the first-arrival amplitude [26, § 4, proof of Thm. 2]. The mean number of calls is $1 + \langle u, Nu\rangle$ [11, § 3, Eq. (8)].

Bourgain et al. state their dimension formula in terms of the smallest reducing subspace containing X [11, Thm. 3.2, Eq. (9)]. In our notation its orthogonal complement is exactly $0 \oplus Y_0$. Indeed, a vector in that complement stays orthogonal to X under every power of U, so it is internal and never emits. Conversely, $0 \oplus Y_0$ reduces U by Theorem 4.1 and is orthogonal to X.

The subspace in their formula is therefore $X \oplus Y_1$. Summing the quadratic form of $I + N$ over an orthonormal basis of X gives $\dim X$ times its uniform unit-sphere average, so the mean call counts sum to $\dim X + \dim Y_1$. Removing the initial call for each basis input gives $\operatorname{tr} N = \dim Y_1$.

Phase invariance and agreement. On a nonempty external type, the phase family also characterizes agreement. If $x_k = b_k L$ for every exit, then $\Theta(z) = f(z)L$ with $f(z) = \sum_k b_k z^k$. For $|z| = 1$ the marked body is unitary, so its trace $\Theta(z)$ preserves the norm of any unit input. Since L does too, $|f(z)| = 1$. This scalar phase cancels in the data channel. Conversely, if those channels are all equal, they equal $\mathscr{E}_L$ because $\Theta(1) = L$. Their average is $\mathscr{M}$ by (8.1), so $\mathscr{M} = \mathscr{E}_L$. Thus agreement says exactly when changing the return phase leaves the data channel unchanged.

F Clause programs and their observations

Two scalar loops in sequence. To write the two copies of `guardRound` from Section 8 as one body, we give each return wire its own basis value. The open matrix acts on $\mathbb{C} \oplus (\mathbb{C} \oplus \mathbb{C})$. Both wires receive the same phase z, so the response in sequence is $f(z)^2$, where $f(z) = (t-z)/(1-tz)$, $t = 3/5$ and $s = 4/5$. To read its columns, let u enter the first rotation and y_1, y_2 enter the two return ports. The first rotation sends $tu - sy_1$ into the second. The external output is therefore $t(tu - sy_1) - sy_2$, and the two return outputs are $su + ty_1$ and $s(tu - sy_1) + ty_2$. Reading off the coefficients of u, y_1, y_2 in turn gives the three clauses.

```
round twoSections : Unit + Qubit = {
  exit unit ↔ [9/25 . exit unit + 4/5 . back 0 + 12/25 . back 1];
  back 0 ↔ [-12/25 . exit unit + 3/5 . back 0 + -16/25 . back 1];
  back 1 ↔ [-4/5 . exit unit + 3/5 . back 1]
}

trace twoSections on unit
```

With no added phase the two sign changes cancel, so the program returns $|*\rangle$ with amplitude $f(1)^2 = (-1)^2 = 1$. Changing the phase to -1 reverses the sign of every odd exit, yet the amplitude is still $f(-1)^2 = 1^2 = 1$. The odd exits must therefore sum to zero and the even exits to one, so a period-two counter reads 0 with certainty.

Shared and separate counts. Measuring each loop can reveal a distinction that the total alone misses. For example, let the first body send $|0\rangle$ straight through and delay $|1\rangle$ by one return. Let the second do the reverse, with unit amplitude on each path. On input $|+\rangle$, separate counters leave the joint state $(|0,1\rangle \otimes |0\rangle + |1,0\rangle \otimes |1\rangle)/\sqrt{2}$. The two records are orthogonal, so tracing them out leaves $I/2$. A shared counter carries $|1\rangle$ on both paths, giving $|1\rangle \otimes |+\rangle$. In the combined body the only nonzero exit map is therefore $x_1 = I$, and measuring its outer tag also preserves $|+\rangle$.

F.1 A flip on every call

In `flipRound`, each call rotates the guard by R_θ and flips the data qubit. Both clauses therefore call `X` before forming their two outputs.

```
X :: Qubit ↔ Qubit
X = { 0 ↔ 1; 1 ↔ 0 }

round flipRound : Qubit + Qubit = {
  exit x ↔ let y = X x in [ 3/5 . exit y + 4/5 . back y];
  back x ↔ let y = X x in [-4/5 . exit y + 3/5 . back y]
}
```

Write $|gd\rangle$ with guard g first (0 for `exit`, 1 for `back`). On `exit 0`, the first clause binds $x = 0$ and the call to `X` gives $y = 1$. Of its two outputs, only $(4/5)|11\rangle$ reaches the second call. This time the `back` clause flips the data to 0, so the part that leaves has amplitude $(4/5)(-4/5) = -16/25$. The returning part is $(12/25)|10\rangle$, so a third call emits $-(48/125)|01\rangle$ and sends $(36/125)|11\rangle$ back. With the guard first, the open matrix is $R_\theta \otimes \mathrm{X}$.

The exit maps of `flipRound`. Put $t = 3/5$ and $s = 4/5$. The two clauses of `flipRound` give $A = t\mathrm{X}$, $C = s\mathrm{X}$ on entry and $B = -s\mathrm{X}$, $D = t\mathrm{X}$ on return. Hence $x_0 = t\mathrm{X}$ and $x_1 = BC = -s^2 I$. The data flips twice before that second exit. In general $x_{k+1} = -s^2 t^k \mathrm{X}^{k+2}$. Grouping equal data gates gives two

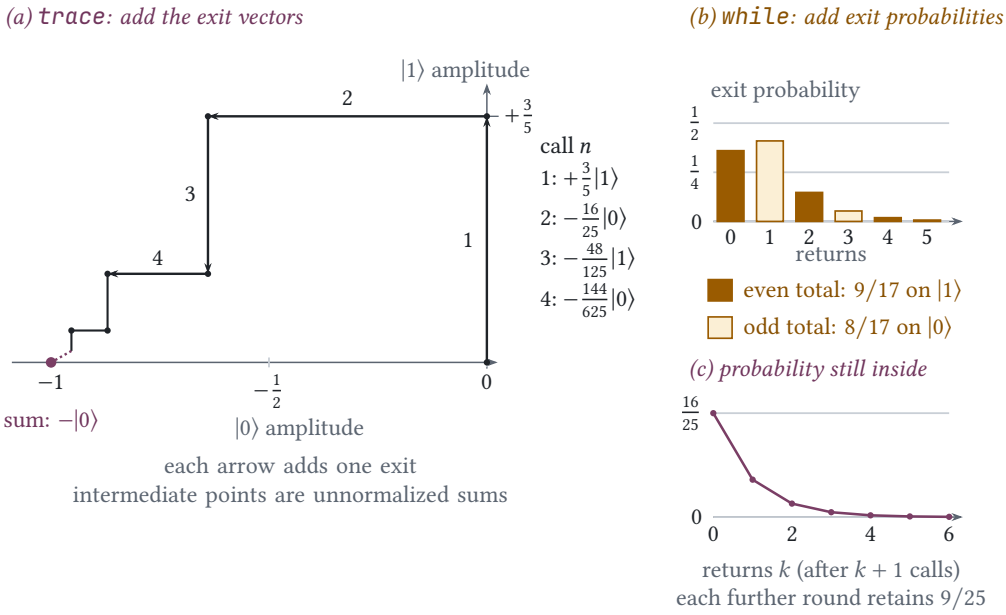


Fig. 19. On input $|0\rangle$, the left panel adds one exit amplitude per call, starting with call one. The bars at right give the exit probabilities by number of returns, starting at zero for the first call. Panel (c) tracks the probability still inside after each call.

ordinary geometric series,

$$\underbrace{t - s^2 t \sum_{j\geq 0} t^{2j} = 0,}_{\text{coefficient of X}} \qquad \underbrace{-\,s^2 \sum_{j\geq 0} t^{2j} = -1,}_{\text{coefficient of } I} \qquad s^2 = 1 - t^2.$$

Thus the whole coherent loop returns $-u$. In particular it can never return data 1 on input 0. With an exact count the exits that return 0 on this input have odd counts, with amplitudes $-s^2, -s^2t^2, \ldots$. Their squared magnitudes sum to $s^4/(1-t^4) = 8/17$. The complementary probability $9/17$ goes to 1. These are the two observations in Figure 19. If we keep only parity, the even counts carry the same label and their amplitudes cancel by the first geometric sum. The odd counts carry the other label and add to -1. Under measurement, exits at different counts contribute separately to the probabilities above.

In Figure 4, eliminating `back 0` leaves a self-loop of amplitude $9/25$ on `back 1`. Input 0 sends $4/5$ there, so successive returns sum to $(4/5)/(1-9/25) = 5/4$. It contributes $(-4/5)(5/4) = -1$ to output 0 and $(-12/25)(5/4) = -3/5$ to output 1, canceling the immediate $3/5$. Interchanging the bits gives the second column of $-I$.

Kickback with a returned ancilla. In `flipRound` the payload changes on every call. We now give the payload two qubits and arrange for the closed loop to flip the data while returning the ancilla. On each call we rotate the guard by $\pi/4$ and apply the same gate M to the payload. Write $c = 1/\sqrt{2}$. The immediate exit is cM, the first delayed exit is $-c^2M^2$, and each further return adds a factor cM. Since $2c^2 = 1$, their sum is $f(M) = cM - c^2M^2(I - cM)^{-1} = M(cI - M)(I - cM)^{-1}$.

Let $J = \mathrm{X}$, $G = HJ$ and $\Pi_{\pm} = (I \pm J)/2$. We choose $M = G \otimes \Pi_{+} + JG \otimes \Pi_{-}$ so that the data control the ancilla in the J basis. We change that basis with H, apply G or JG to the ancilla according to the resulting data bit, then undo the basis change.

Since $G|0\rangle = |-\rangle$ and $G|1\rangle = |+\rangle$, expanding $\Pi_{\pm} = (I \pm J)/2$ gives $M = |+\rangle\langle 1| \otimes I + |-\rangle\langle 0| \otimes J$. On the two payload bits this reads

$$M|0, d\rangle = \frac{|0, 1-d\rangle - |1, 1-d\rangle}{\sqrt{2}}, \qquad M|1, d\rangle = \frac{|0, d\rangle + |1, d\rangle}{\sqrt{2}}.$$

The data bit flips in the first case and stays in the second. Each nonzero coefficient of M is $\pm 1/\sqrt{2}$, and the guard contributes another $1/\sqrt{2}$, leaving only coefficients $\pm 1/2$. The listing can therefore use exact rational coefficients.

```
round kickbackExact : (Qubit x Qubit) + (Qubit x Qubit) = {
  exit ⟨0, 0⟩ ↔ [1/2 . exit ⟨0, 1⟩ + -1/2 . exit ⟨1, 1⟩
                 + 1/2 . back ⟨0, 1⟩ + -1/2 . back ⟨1, 1⟩];
  exit ⟨0, 1⟩ ↔ [1/2 . exit ⟨0, 0⟩ + -1/2 . exit ⟨1, 0⟩
                 + 1/2 . back ⟨0, 0⟩ + -1/2 . back ⟨1, 0⟩];
  exit ⟨1, 0⟩ ↔ [1/2 . exit ⟨0, 0⟩ + 1/2 . exit ⟨1, 0⟩
                 + 1/2 . back ⟨0, 0⟩ + 1/2 . back ⟨1, 0⟩];
  exit ⟨1, 1⟩ ↔ [1/2 . exit ⟨0, 1⟩ + 1/2 . exit ⟨1, 1⟩
                 + 1/2 . back ⟨0, 1⟩ + 1/2 . back ⟨1, 1⟩];
  back ⟨0, 0⟩ ↔ [-1/2 . exit ⟨0, 1⟩ + 1/2 . exit ⟨1, 1⟩
                 + 1/2 . back ⟨0, 1⟩ + -1/2 . back ⟨1, 1⟩];
  back ⟨0, 1⟩ ↔ [-1/2 . exit ⟨0, 0⟩ + 1/2 . exit ⟨1, 0⟩
                 + 1/2 . back ⟨0, 0⟩ + -1/2 . back ⟨1, 0⟩];
  back ⟨1, 0⟩ ↔ [-1/2 . exit ⟨0, 0⟩ + -1/2 . exit ⟨1, 0⟩
                 + 1/2 . back ⟨0, 0⟩ + 1/2 . back ⟨1, 0⟩];
  back ⟨1, 1⟩ ↔ [-1/2 . exit ⟨0, 1⟩ + -1/2 . exit ⟨1, 1⟩
                 + 1/2 . back ⟨0, 1⟩ + 1/2 . back ⟨1, 1⟩]
}

trace kickbackExact on 3/5 . ⟨0, 0⟩ + 4/5 . ⟨1, 1⟩
```

To see which qubit receives the sign, look at $f(M)$ in each sector. For $M = G$, the identity $G^2 = 2cG - I$ gives $G(cI - G) = I - cG$. The numerator cancels the denominator, so $f(G) = I$. For $M = JG$, we have $(JG)^2 = I$, so the numerator is now $cJG - I = -(I - cJG)$ and $f(JG) = -I$. Both denominators are invertible since $c < 1$ and both gates are unitary. Since the ancilla comes back in either sector, the relative sign acts on the data, giving $I \otimes (\Pi_{+} - \Pi_{-}) = I \otimes J$ on every joint input.

The input is entangled between ancilla and data. Its output is $3|0, 1\rangle/5 + 4|1, 0\rangle/5$, exactly the result of applying J to the second component. The same identity on arbitrary joint inputs follows from the two-sector calculation above.

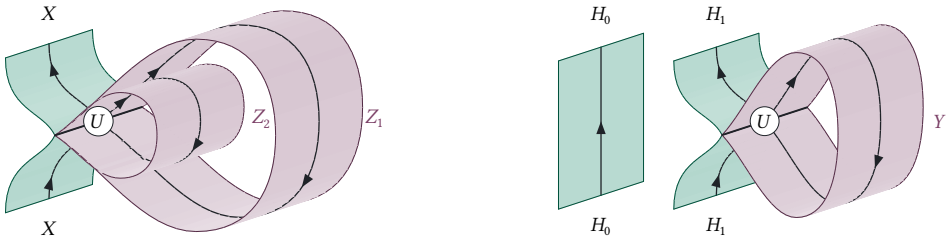
G Reading the feedback diagrams

Read sheets from bottom to top. A summand occupies a surface, its tensor factors run along the depth, and the seam U acts on both surfaces even where one hides the other [17, Defs. 16, 18–20, 22]. Bending the return surface closes the summand Y of $X \oplus Y$. Hiding a register instead takes the partial trace over its tensor factor.

The finite trace satisfies the trace laws even for a body with a trapped return [3, Cor. 3.3.16] [6, Thms. 7–8]. The measured loop in Appendix C has a least fixed point. For the coherent loop, deriving

a fixed-point operator from the trace by the cartesian construction would require a diagonal [27, Thm. 3.1]. For direct sum, $u \mapsto (u, u)$ doubles the squared norm of every nonzero u, so it cannot be a unitary in our language.

Figure 20 shows two ways to use the trace laws: close two return ports together, or close one beside a bypass. In each equation we regroup the summands to put the return ports last.



$$\mathrm{Tr}^{Z_1}(\mathrm{Tr}^{Z_2} U) = \mathrm{Tr}^{Z_1 \oplus Z_2} U \qquad I \oplus \mathrm{Tr}^{Y} U = \mathrm{Tr}^{Y}(I \oplus U)$$

Fig. 20. At left, closing Z_2 and then Z_1 gives the same gate as closing $Z_1 \oplus Z_2$ together. At right, the external space is $H_0 \oplus H_1$: H_0 passes straight through and U acts on $H_1 \oplus Y$, with Y fed back.

The right sheet gives the block form used for `bypassRound`. The bypass contributes only to the immediate exit, whose block is $I \oplus x_0$. Each later exit is $0 \oplus x_k$, so their sum is $I \oplus L$. On the left, closing the two ports in either order solves both wire equations, as in Proposition 4.2.

G.1 The backward wire in the Int construction

To type a backward wire, Int pairs a forward type P^+ with P^-. A map from P to Q takes P^+ from the left and Q^- from the right, giving the underlying unitary $P^+ \oplus Q^- \to Q^+ \oplus P^-$ in Figure 21.

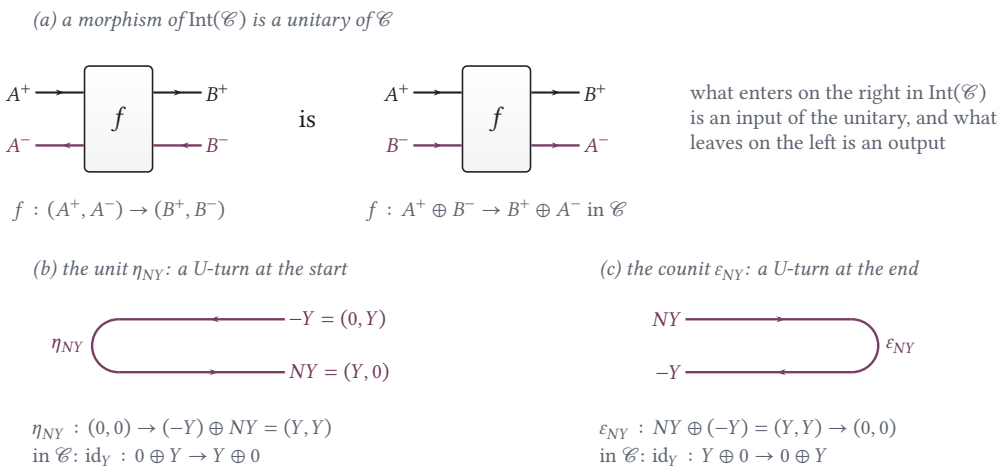


Fig. 21. Objects in the compact completion have forward and backward components. Cups and caps bend a port within that completion.

In Figure 22, order both inputs and outputs rightward port first. Then f sends $b^+ = f_{11}a^+ + f_{12}b^-$ to g, which returns $b^- = g_{21}b^+ + g_{22}c^-$. Substituting gives

$$b^+ = \underbrace{f_{11}a^+ + f_{12}g_{22}c^-}_{\text{from outside}} + \underbrace{f_{12}g_{21}b^+}_{\text{one trip past the cut}} .$$

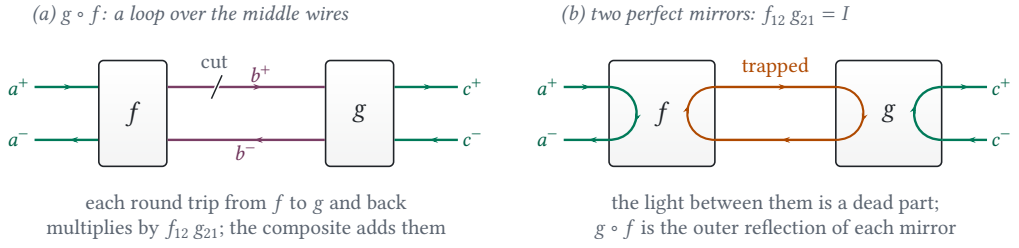


Fig. 22. Composition closes the middle ports. Between two perfect mirrors, a round trip returns the light unchanged and never reaches an outer port.

This is the wire equation of Section 4, with $y = b^+$ and $D = f_{12}g_{21}$. Once we solve it, f and g give the external outputs a^- and c^+. When $I - D$ is invertible this elimination gives the scattering matrix of the pair, the Redheffer star product [24, Figs. 10–11].

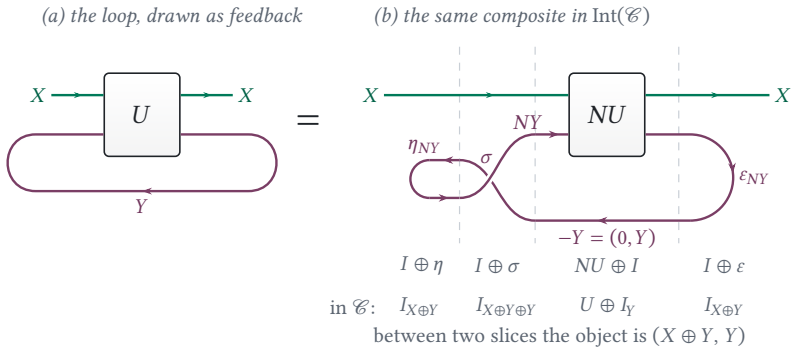


Fig. 23. Both diagrams close Y in $U : X \oplus Y \to X \oplus Y$. The cap takes the return onto the backward wire $(0, Y)$ and the cup brings it to the input. Each entry in the lower row gives that slice's underlying unitary before composition in Int($\mathscr{C}$).

The fully faithful embedding $N(X) = (X, 0)$ gives X no backward port. Bending Y around $N(U)$ identifies its returning output $Cu + Dy$ with its returning input y. Thus the compact closed trace in Figure 23 gives $N(L)$ by Theorem 4.1 [35] [39, Thms. 5.15–5.16].

In the sum of P and Q, the forward ports occur in the order $P^+ \oplus Q^+$ and the backward ports in the reverse order $Q^- \oplus P^-$. To close the Y wire, the counit bends the forward port $(Y, 0)$ onto its dual $(0, Y)$ and the unit bends it forward again. Both bends carry Y unchanged, so composing them with $N(U)$ joins the ports by the original trace. With $\eta_{NY} : (0, 0) \to (-Y) \oplus N(Y)$ and $\varepsilon_{NY} : N(Y) \oplus (-Y) \to (0, 0)$, the four maps compose as follows

$$N(\mathrm{Tr}^Y U) = (I_{NX'} \oplus \varepsilon_{NY})(N(U) \oplus I_{-Y})(I_{NX} \oplus \sigma_{-Y,NY})(I_{NX} \oplus \eta_{NY}).$$

Figure 23(b) draws the case $X' = X$. Read its four regions from left to right. The cup opens the two ports and the swap puts them in the order $N(U) \oplus I_{-Y}$ expects. In each direction Y moves past the zero space, so the underlying swap is $I_{Y\oplus Y}$, or $I_{X\oplus Y\oplus Y}$ with the external wire. The cap then joins the ports. Both bends belong to the compact completion.

H A ring realization of the same exits

In Figure 6, light stays in its waveguide with amplitude $t = 3/5$ and crosses to the other waveguide with amplitude $is = 4i/5$ [29, Eq. (1)]. The light field extends a little beyond each guide, so the fields

overlap across the narrow gap [2]. Light crossing into the ring from u keeps its direction of travel. It leaves the gap as d, comes around again as y, and the part that crosses back joins the direct light on its way to v.

We can see where the i comes from by checking the columns, just as we did for `guardRound`. With through amplitude $t = 3/5$ and the same crossing amplitude x in both directions, the columns are (t, x) and (x, t). Their inner product $t(x + \bar{x})$ must vanish, so x is purely imaginary. Here we choose $x = is$. This coupler is the rotation from `guardRound` in a different basis on the return wire, $(1 \oplus i)R_\theta(1 \oplus (-i))$. To leave after one trip, light crosses into the ring with amplitude is and crosses back with amplitude is. Each further trip adds a factor t for staying in the ring at the coupler. The exit after $k + 1$ trips thus has coefficient $(is)t^k(is) = -s^2t^k$. Hence the coupler and the rotation have the same exit on every round and the same response $\Theta_{\text{ring}}(z) = (t - z)/(1 - tz)$ at round-trip phase z. In code, `(a :+ b)` denotes $a + bi$. Chaining the scaling dots multiplies the coefficients, giving $i\,s$ below.

```
round ringCoupler : Unit + Unit = {
  exit unit ↔ [3/5 . exit unit + (0 :+ 1) . 4/5 . back unit];
  back unit ↔ [(0 :+ 1) . 4/5 . exit unit + 3/5 . back unit]
}

trace ringCoupler on unit
```

With no added phase the ring returns $-|*\rangle$. If instead a trip preserves only a fraction a of the field amplitude, each return carries az and the response is $(t - az)/(1 - taz)$ [2, Eq. (6)]. At $z = 1$ its difference from -1 is $(1 + t)(1 - a)/(1 - ta)$. Thus even small losses matter near $t = 1$. When $a = t < 1$, the direct and returning fields cancel exactly.

For a finite body with the same attenuation a on every return, set $z = w = a$ in the inner-product identity of Appendix E. On unit input u this gives $1 - \|\Theta(a)u\|^2 = (1 - a^2)\|F_a u\|^2$. Thus the circulating weight also measures sensitivity to a small loss: for $a = 1 - \epsilon$ the output loses $2\epsilon\langle u, Q(1)u\rangle + O(\epsilon^2)$ of its intensity at zero return phase. For a fixed body and a small common phase error δ, the amplitude changes to first order by $i\delta LQ(1)$, since the complex derivative at $z = 1$ is $\Theta'(1) = LQ(1)$.

To read the two right panels of Figure 8 from this program, we also need to say when each exit reaches the straight guide. Write $u(T)$ for the incident field, with time measured relative to the direct path. The copy that makes k trips arrives $k\tau_0$ later with the amplitude of exit k. The field at the marked port is therefore

$$v(T) = t\,u(T) - s^2 \sum_{k \geq 1} t^{k-1} u(T - k\tau_0).$$

Raymer and McKinstrie give this train of delayed copies with the opposite overall output sign [46, Eqs. (30)–(33)]. If the pulse ends before the first return, the delayed copies arrive separately. For a steady wave that completes a whole number of cycles in time τ_0, shifting it by any multiple of τ_0 gives back $u(T)$. Their coefficients sum to $-s^2/(1 - t) = -8/5$. Adding the direct $3/5$ gives the sign returned by the program [2, Eqs. (6)–(7)].

To realize `bypassRound`, match the bypass to the straight-through path in phase and travel time. Prepare the external qubit in $|+\rangle$, run the two paths, and apply H to compare them. At $z = 1$ the ring changes the sign on its path while the bypass preserves it, so the difference port has probability $|1 - (-1)|^2/4 = 1$.

For a short pulse, only the direct field arrives with the bypass. Their amplitudes at the difference port are $-t/2$ and $1/2$, so the first bin contributes $(1 - t)^2/4$. Each later bin contains just one delayed

exit. The two splitters give its amplitude a factor $1/2$, and we add the resulting probabilities.

$$\Pr(b=1)=\frac{1}{4}\left((1-t)^2+\sum_{k\ge 1}s^4t^{2(k-1)}\right)=\frac{1}{4}\big((1-t)^2+1-t^2\big)=\frac{1-t}{2}=\frac{1}{5}.$$

The bypass thus makes the sign visible in steady light. When only the direct pulse meets the bypass, the arrival bins give the measured row of Table 1.

A pulse contains a range of frequencies, each with its own round-trip phase $\omega\tau_0$. Suppose the incident photon has the same path state $|+\rangle$ at every frequency, with frequency distribution $p(\omega)$. We count arrivals at the difference port over the entire pulse. Since distinct frequency modes are orthogonal, ignoring frequency adds their probabilities, weighted by $p(\omega)$, giving $\int p(\omega)|1-\Theta_{\text{ring}}(e^{i\omega\tau_0})|^2\,d\omega/4$. The component at resonance goes entirely to the difference port. For a component whose round-trip phase is δ away from a whole turn, the probability of reaching the other port, for fixed $|t|<1$, is

$$\frac{|1+\Theta_{\text{ring}}(e^{i\delta})|^2}{4}=\frac{(1+t)^2\sin^2(\delta/2)}{(1-t)^2+4t\sin^2(\delta/2)}=\frac{Q(1)^2}{4}\delta^2+O_t(\delta^4).$$

If the spectrum lies in $|\delta|\le h$ and $\sigma^2=\mathbb{E}[\delta^2]$, then $\mathbb{E}[\delta^4]\le h^2\sigma^2$. As $h\to 0$ the mean probability of reaching the other port is therefore $Q(1)^2\sigma^2/4+o(\sigma^2)$. If the round-trip phase is uniform, averaging the full response over the spectrum gives the measured loop by (8.1).

When we first turn on the light, the output after n trips still differs from -1 by $(1+t)t^n$ (Section 5). Each trip multiplies this difference by t. Near $t=1$ it takes about $-1/\log t\sim 1/(1-t)$ trips to reduce it by a factor of e.

Once the output is steady, we can vary the frequency and read the change in its phase. With a frequency-independent coupler, only the phase $\varphi(\omega)$ of a round trip changes. The chain rule gives $-i\Theta^\dagger d\Theta/d\omega=(d\varphi/d\omega)Q(e^{i\varphi})$ [50, 56]. In a nondispersive guide $\varphi=\omega\tau_0$, so at resonance this is $\tau_0 Q(1)$, where $Q(1)$ is the circulating intensity for unit input [29, Eq. (7)]. For a wave packet, the corresponding dwell time involves averaging the Wigner–Smith matrix over the packet's energy spectrum [21, Eqs. (34)–(35)].

I Observing the response

To put phase z on each return we compose the body with $I_b\oplus P_z^c$, where $P_z^c=\{s\leftrightarrow z\cdot s\}_{s\in\mathscr{B}(c)}$ and $|z|=1$. With the bypass the program is

$$H;\mathsf{trace}_1\big(\texttt{bypassRound};(I_{\texttt{Qubit}}\oplus P_z^1)\big);H.$$

Only the rotation returns, once on average, so $N_{\text{bypass}}=\text{diag}(0,1)$ gives mean $1/2$ on $|+\rangle$. The final Hadamard gives $\Pr(b=1)=|1-\Theta_{\text{rot}}(z)|^2/4$. Averaging over the circle gives $1/5$ by (8.1).

φ	0	$\pi/2$	π
$\Theta_{\text{rot}}(e^{i\varphi})$	-1	$(15-8i)/17$	1
$\Pr(b=1)$	1	$1/17$	0

Comparing two nearby phases. The bypass compares the loop with a straight path. To compare its weights 4 and $1/4$ (return amplitudes 2 and $1/2$), we instead control two nearby phase settings with a qubit $H_r=\mathbb{C}^2$. Distributivity puts that qubit on both ports of the body. For $P_\delta=\text{diag}(1,e^{i\delta})$ the controlled body is

$$W_{z,\delta}=\begin{pmatrix}I_{H_r}\otimes A & I_{H_r}\otimes B\\ P_\delta\otimes zC & P_\delta\otimes zD\end{pmatrix}.$$

On the two reference branches exit k acquires z^k and $(ze^{i\delta})^k$, respectively. Summing gives the controlled gate $|0\rangle\langle 0|_r \otimes \Theta(z) + |1\rangle\langle 1|_r \otimes \Theta(ze^{i\delta})$. On $|+\rangle \otimes u$ with $\|u\| = 1$ this gives the state in Figure 11.

After $S^\dagger$ and H, the data at outcomes zero and one are $(\Theta(z)u \mp i\Theta(ze^{i\delta})u)/2$. Subtracting their squared norms cancels the branch norms. The remaining imaginary overlap $\mu = \mathrm{Im}\langle \Theta(z)u, \Theta(ze^{i\delta})u\rangle$ is $\delta\langle u, Q(z)u\rangle + O(\delta^2)$ by Theorem 8.1.

Scoring zero as $+1$ and one as -1 gives mean μ and variance $1 - \mu^2$. Over n independent runs the mean score divided by δ has variance $(1-\mu^2)/(n\delta^2)$ and bias $O(\delta)$. At a fixed body and phase, taking δ proportional to ϵ and n proportional to ϵ^{-4} gives bias and standard error $O(\epsilon)$. Alternatively, subtracting the imbalances at opposite offsets and dividing by 2δ cancels even powers and gives bias $O(\delta^2)$.

At resonance the scalar rotation gives an exact check on the step size. Using the scalar response in the overlap gives

$$\frac{\mu}{\delta} = \frac{1+t}{1-t}\,\frac{\sin \delta/\delta}{1 + 4t\sin^2(\delta/2)/(1-t)^2}.$$

The first factor is the weight. For the estimate to approach it with vanishing relative error as $t \to 1^-$, the phase step must shrink faster than $1 - t$.

For phases 1 and i, `guardRound` returns -1 and $(15 - 8i)/17$. Their overlap $(-15 + 8i)/17$ gives a probability difference of $8/17$, so the caller reads zero with probability $25/34$ and one with $9/34$. A quarter turn is a large phase change. Dividing the measured difference by $\pi/2$ gives $16/(17\pi)$ for this setting. To read the local weight, we repeat the comparison with smaller angles δ. The probability difference shrinks, but its ratio to δ approaches four.